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Persistent Dealer-Gamma Regimes Under Temporal Obfuscation: A Re-Run on Complete Inputs with a Controlled Large Language Model Re-Test

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Abstract: Temporal obfuscation tests whether a large language model (LLM) reasons about numbers or recalls market history. Our earlier work reported that an LLM detects persistent dealer-gamma regimes in 30-day windows of obfuscated daily gamma exposure (GEX) of SPY, with a sharp rise in detections after 2023. Here we re-run that test on the complete data. The published 2024–2025 inputs were the call leg of the options chain without the put leg. We rebuilt six daily series, replayed the deterministic rule on them, and bridged o4-mini to gpt-6-luna on the exact published inputs. We also reran the 2024 and 2020 baselines with relative day labels instead of calendar dates. On the published inputs the rule detects 501 of 501 windows (100.0%) of 2024–2025; on the complete series it detects 60 of 502 (12.0%), and gpt-6-luna 63 of 502 (12.5%). The two models agree on 99.6% of the published windows (kappa 0.993), and gpt-6-luna follows the rule in 99.7% of 4,671 windows (kappa 0.986). Removing same-day expiries gives 14.7%, not the published rate. With relative labels gpt-6-luna detects 80.7% of the 2024 baseline windows, against 81.2% with dates. The complete series shows no post-2023 increase that the data distinguish from zero.

Keywords: LLM validation; regime detection; gamma exposure; 0DTE options; temporal obfuscation; data provenance; reproducibility

1. Introduction

A language model that labels a market regime may be recalling the period it was trained on, or it may be reasoning about the numbers it is shown. Temporal obfuscation separates the two readings: dates and tickers are removed, so that the model has only the values and the stated criteria. The test is only as sound as the numerical series that it shows to the model. This article re-runs the published 30-day regime test on the complete daily series, with a second model on identical inputs, and asks what the published conclusions rest on.

1.1. Motivation: Data-Definition Drift and Model Retirement

The published run used o4-mini in January 2026. The model no longer answers inference requests (the endpoint returns `model_not_found`). The raw options table was reloaded on 2026-02-03, and the daily GEX table was rebuilt on 2026-10-08 from stored per-contract gamma, with same-day expiries included (Section 3.3). A result that depends on a model and on an input definition that can both disappear cannot be re-run, only re-interpreted. The study therefore recovers the exact inputs of the published run from the stored batch file, bridges a second model (gpt-6-luna) to o4-mini on these inputs, and stores every input, prompt and response.

1.2. Relation to Prior Work

This article is the extended version of the AIAI 2026 conference paper [1], which introduced 30-day regime detection under temporal obfuscation and built on the single-day study presented at IEEE BigData 2025 [2]. Both lines of work are combined in a separate journal article [3]. The experiments reported here are new relative to all three publications. New relative to the three publications are (i) the provenance and definition of the daily input series, (ii) a second model, gpt-6-luna, bridged to o4-mini on identical inputs, (iii) the years 2008–2019, QQQ and IWM, 2026 and an intraday check, (iv) overlap-aware inference, (v) an obfuscated-versus-dated rerun of the Phase 3 and 4 baselines, and (vi) a disclosed per-symbol scaling for QQQ and IWM. The recap of the earlier design is limited to Section 3.

1.3. Research Questions

We address seven questions:

RQ1. Which daily series were the published inputs, and how do the deterministic labels change on the complete series?
RQ2. Does gpt-6-luna reproduce the o4-mini labels on the exact published inputs?
RQ3. Does the model's regime call follow the input series and the stated criteria?
RQ4. How much of the persistence signal lives in same-day-expiry (0DTE) gamma?
RQ5. Do the answers hold for 2008–2019, for QQQ and IWM, and in the post-publication period (2026)?
RQ6. Does a late-session intraday snapshot reproduce the end-of-day label?
RQ7. Does the 2024-versus-2020 contrast of the earlier baseline runs, whose prompts carried calendar dates, persist when the dates are replaced by relative day labels?

1.4. Contributions

This article makes six contributions. (i) It reconstructs the published inputs and gives an account of the data they carried. The published 2024 and 2025 values are the call leg of the options chain without the put leg, from 2023-09-29 on. The deterministic labels are recomputed on the complete series: the rule detects 501 of 501 windows (100.0%) of 2024 and 2025 on the published inputs and 60 of 502 (12.0%) on the complete series (Section 4.1, Section 4.2). (ii) It analyses the sensitivity of the deterministic labels to the series definition across six daily series (one of them a call-only diagnostic), three assets and 2008–2026. Removing same-day expiries does not restore the published rates (14.7% against 12.0%), and the fixed thresholds leave QQQ and IWM without a detection until a disclosed scaling lifts the rule rate over 2020–2026 to 21.0% (QQQ, 350 of 1,670) and 53.9% (IWM, 901 of 1,671) (Sections 4.5–4.7). (iii) It bridges o4-mini and gpt-6-luna on identical inputs. The two models give the same verdict in 99.6% of the 1,412 published windows (kappa 0.993). On the complete series gpt-6-luna labels 613 of 4,671 windows (13.1%) as persistent and agrees with the rule on 99.7% of the 4,671 windows (kappa 0.986) (Sections 4.3, 4.4 and 4.10). (iv) It applies overlap-aware statistics to stride-one windows: the quarter-block bootstrap and a non-overlapping subsample next to the Wilson intervals. The pooled rule rate of 12.0% on the complete series has a Wilson interval of 9.4–15.1% and a bootstrap interval of 5.0–19.2% (Section 3.8). (v) It stores every input, prompt and response in result tables (36,634 answers), from which each table and figure of this article is generated (Section 3.7). (vi) It reruns the Phase 3 and 4 baselines with relative day labels instead of calendar dates. For 2024, gpt-6-luna detects 180 of 223 windows (80.7%) with relative labels and 181 of 223 (81.2%) with dates, so the visible dates changed nothing that the data can detect (Section 4.11).

1.5. Organization

Section 2 reviews dealer gamma exposure, the validation of LLM reasoning by obfuscation and the construct validity of financial inputs. Section 3 defines the daily series, the windows and classifier,

the model protocol, the experimental arms and the statistics. Section 4 reports the results, from the lineage of the published inputs to the rerun of the earlier baselines. Section 5 discusses what survives, what the model tracks, the threats to validity and the implications. Section 6 concludes, and the appendices give the prompt, schema and API configuration (Appendix A), the series definitions in SQL (Appendix B) and the per-year tables (Appendix C).

2. Background and Related Work

2.1. Dealer Gamma Exposure and 0DTE Options

Market makers must keep their books close to delta-neutral, so accumulated gamma turns into systematic hedging flows [4,5]. Dealers are the counterparties of customer option positions, and aggregated market makers carry negative gamma that feeds back to the spot market [6]. Dealer hedging has measurable price effects [7,8], and practitioners summarise it as a gamma exposure (GEX) figure [6,9]. Such a figure is a lossy scalar: it depends on modelling choices (which gamma, which contracts, which expiries), so two daily GEX series with the same name need not agree.

The growth of zero-days-to-expiration (0DTE) options since 2022 concentrates gamma at very short maturities [10–12]. Whether 0DTE contracts are included is one such modelling choice, and it can change the sign of the daily figure on some days. Regime-switching models detect regimes from observable outcomes such as returns [13–15]. The rule and the LLM in this paper work on dealer positioning instead.

2.2. Validating LLM Reasoning via Obfuscation

Concern that LLM performance reflects memorized rather than inferred structure [16–18] motivates behavioral tests of the CheckList type [19]. Temporal obfuscation removes dates, tickers and event context so that only numerical structure remains; it is one such test [2]. A model that still gives the same answer on obfuscated input is less likely to be recalling training data [17].

Obfuscation has a limited reach. It tests whether the answer depends on memorized context. It does not test whether the numbers shown to the model mean what the prompt says they mean. A model can pass an obfuscation test and still report faithfully on a flawed input. That second question concerns the inputs, not the model, and is the subject of the next subsection.

2.3. Construct Validity of Financial Inputs

An LLM validation inherits the construct validity of its inputs [20]. A daily GEX series is a derived quantity. Its definition (gamma model, contract filters, 0DTE treatment, put and call legs) and its provenance (vendor, load date, rebuild date) are part of the experiment, and they belong in the record of a study as much as the prompt and the model name do.

Three threats follow for the reproducibility of LLM studies. Data definitions drift between collections. Upstream data can be rebuilt silently, so that the same query returns a different series later. Models are deprecated, so a run cannot be repeated with the same model. Section 3 describes how this paper records the input series and the model version for every arm.

3. Methods

The study cuts six daily GEX series and the derived inputs of the rerun arms (38,936 stored daily rows) into 38,501 windows of 30 trading days. Each window is labelled by the deterministic rule and, for each of 18 arms, by a language model; the 36,634 stored answers (1,922 of them backfilled from earlier o4-mini runs) and all other results are kept in parquet files and in the tables `backups.gex_llm_regime_*`. Figure 1 gives an overview. The daily series are defined in Section 3.2 and their lineage in Section 3.3; windows and the classifier follow in Section 3.4, the model protocol in Section 3.5 and the experimental arms in Section 3.6.

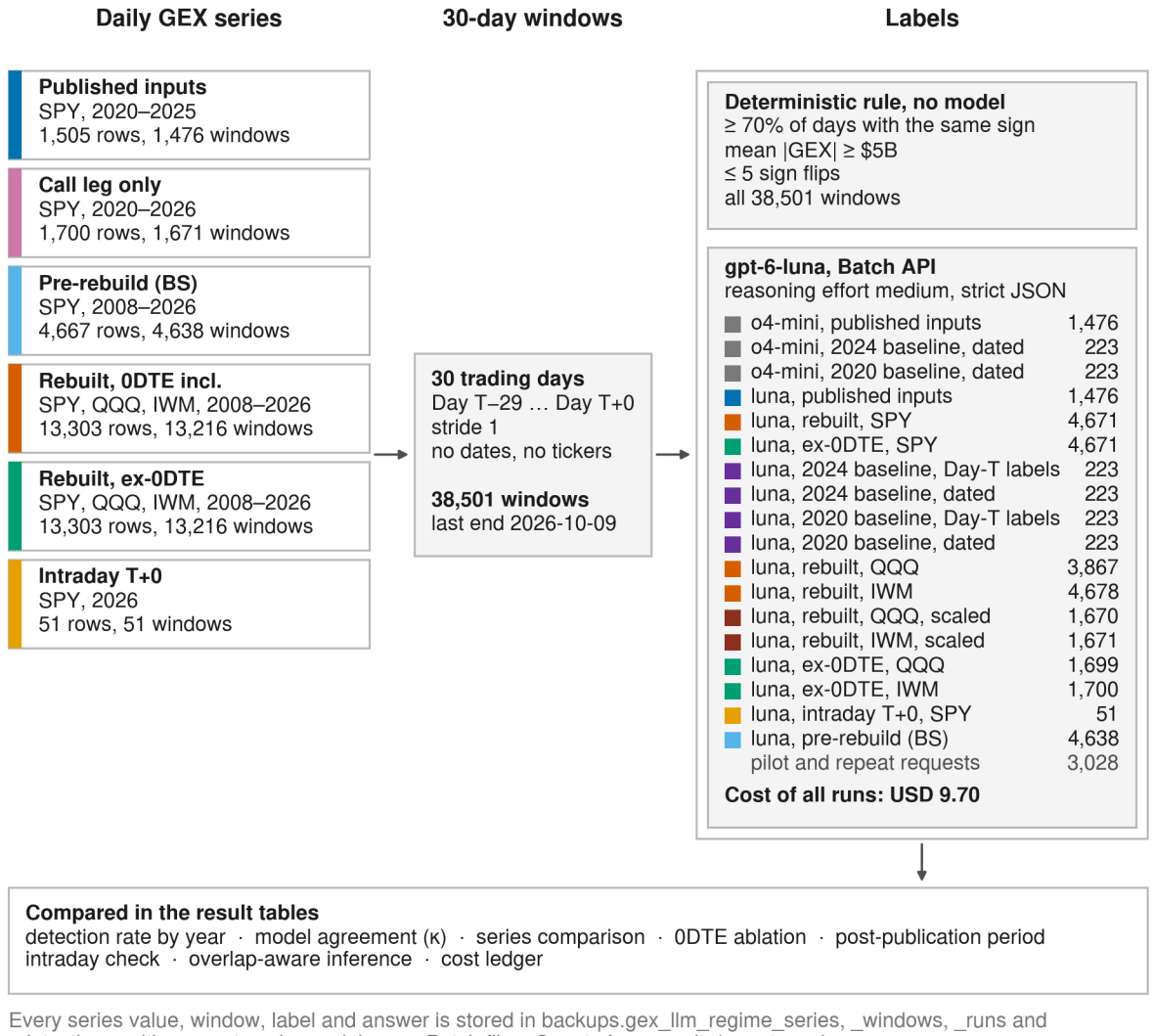


Figure 1. Study design. Six daily GEX series, one of them the call-only diagnostic, and the derived inputs of the rerun arms (Table 1) are cut into 30-trading-day windows, labelled by the deterministic classifier and, for each arm of Table 3, by gpt-6-luna through the Batch API. Boxes give the stored counts, and every input, label and response is stored (Section 3.7). **Read this figure as:** daily rows become 38,501 windows, every window carries a label from the rule, and each of the 36,634 stored answers is compared with the label of its own window.

3.1. Regime Criteria

A 30-trading-day window is a persistent regime when three criteria hold [1,3]: (i) persistence, at least 21 of the 30 days (70%) share the sign of net GEX; (ii) magnitude, the mean absolute net GEX is at least $5 billion; and (iii) stability, the sign changes at most five times. The classifier labels every other window as low conviction (persistent sign and mean magnitude of at least $3 billion, but failing a criterion) or transitional. The detection label used throughout is the binary persistent or not-persistent verdict (`regime_detected` for the model, `clf_should_detect` for the classifier). The thresholds are those of the earlier studies and are not re-tuned. Zero GEX counts as non-positive. A window with a persistent sign and a mean magnitude of at least $5 billion but more than five sign changes fails the stability criterion and is labelled low conviction, because its mean still exceeds $3 billion. The prompt shows values rounded to 0.01B, whereas the rule replays the unrounded USD values stored with each window. The thresholds are fixed in dollars before the runs, for every symbol. This choice has a

consequence for QQQ and IWM, whose gamma is smaller than that of SPY, and Section 4.7 reports the resulting floor effect.

3.2. Daily GEX Series

For each symbol and trading day t the net dealer gamma exposure is

$$\mathrm{GEX}_t = \sum_{i \in \mathcal{C}_t} s_i \, \Gamma_i \, \mathrm{OI}_i \cdot 100 \cdot S_t^2 \cdot 0.01, \qquad s_i = \begin{cases} +1 & \text{call}, \\ -1 & \text{put}, \end{cases} \tag{1}$$

where $\mathcal{C}_t$ is the set of contracts listed on day t with a defined gamma Γ_i, positive open interest OI_i and a positive underlying price S_t. The series of Table 1 differ in how Γ_i is obtained, in which expiries and legs enter $\mathcal{C}_t$, and in where the values come from.

Table 1. Daily GEX series. The first block holds the series of the published result and its diagnostics, the second the complete series, the third the intraday check, the fourth the derived inputs of the rerun arms (Section 3.6).

Series key	Source and gamma convention	Symbols and span	Role
`jan2026_phase4a`	Phase 4A batch file of 2026-01-04: the exact inputs sent to o4-mini, at 0.01B precision	SPY, 2020-01-02 to 2025-12-31	Bridge to the published result
`call_only`	Raw `gex.options_chains_partitioned`: call leg of every contract, same-day expiries included, put leg dropped	SPY, 2020 to 2026-10-09	Reproduces the put-less inputs; no model arm
`old_bs`	Archive `backups.options_daily_summary_pre_rebuild_20261004`: Black–Scholes gamma on integer day counts, same-day expiries zeroed; put-less from about October 2023	SPY, 2008-01-02 to 2026-10-02, with holes	Diagnostic bridge to the pre-rebuild table
`new_full`	`gex.options_daily_summary` after the 2026-10-08 rebuild: stored per-contract gamma, same-day expiries included, both legs	SPY, QQQ, IWM; 2008 (QQQ 2011) to 2026-10-09	Primary complete series
`new_ex0dte`	Raw `gex.options_chains_partitioned`, contracts with `expiration > trading_date`; otherwise as `new_full`	As `new_full`	0DTE ablation
`intraday_late`	`gex.intraday_gex_summary`, snapshot type `final_rush` (19:55:59 UTC), stored gamma	SPY, 2026-07-27 to 2026-10-09	Intraday check
`new_full_scaled`	`new_full` of QQQ and IWM times one fixed factor per symbol: the median, from 2020-01-01 to 2026-10-09, of SPY's open-interest notional divided by the symbol's (`total_weighted_oi` × 100 × underlying price, from `gex.options_daily_summary`)	QQQ, IWM; 2020 to 2026-10-09	Scaled cross-asset arms (protocol change)
`p3_2024`	Phase 3 batch file of 2025-11-20: 252 daily values, gross call-plus-put magnitude without same-day expiries and with a negative sign imposed (negative on 242 days; the ten positive values are placeholder rows of the source database)	SPY, 2024-01-02 to 2024-12-31	2024 baseline, Day-T labels
`p3_2024_dated`	The same values with the ISO dates that were sent in November 2025	As `p3_2024`	2024 baseline, dated labels
`p4_2020`	Phase 4 batch file of 2025-11-20: 252 daily values, magnitude only (positive on every day)	SPY, 2020-01-02 to 2020-12-31	2020 baseline, Day-T labels
`p4_2020_dated`	The same values with the ISO dates that were sent in November 2025	As `p4_2020`	2020 baseline, dated labels

Each series answers one question. `jan2026_phase4a` holds the numbers that o4-mini saw in January 2026, so it is the only input that can link the new runs to the published rates. `call_only` is rebuilt from the raw chain with the call leg alone and tests whether the put-less inputs alone explain the published rates. `old_bs` is the pre-rebuild daily table and shows how far a Black–Scholes gamma on integer day counts moves the rates. `new_full` is the complete series and the primary object of the study. `new_ex0dte` removes same-day expiries and so isolates their contribution. `intraday_late` supplies a late-session value for the last day of a window. The fourth block serves the cross-asset

and baseline replays. `new_full_scaled` rescales QQQ by the factor 4.22 (over 1,699 days) and IWM by 11.43 (over 1,700 days), so that the fixed dollar thresholds are comparable across the three ETFs. The Phase 3 file carries a negative sign on 242 of its 252 days (the ten positive values are placeholder rows), the Phase 4 file is positive on all 252 days, and the dated twins differ from their obfuscated counterparts in the day labels only.

Windows use the rows of each series (rule A, Section 3.4), so the calendars matter. The series `old_bs` lacks SPY on 2023-07-12, 2024-03-11 and on the 26 trading days from 2026-01-02 to 2026-02-09, and the January calendar lacks 2023-07-12 and 2024-03-11; both dates are in the complete series. For the intraday series, the `market_close` slot (20:00:59 UTC) equals the end-of-day row on 51 of 54 days, so the independent late-session observation is `final_rush`. The irregular days 2026-07-29, 2026-08-03 and 2026-09-14 are excluded.

3.3. Provenance of the Published Inputs: Put-Row Loss, Reload and Rebuild

The published inputs cannot be regenerated from today's database. The raw options table was reloaded on 2026-02-03, one month after the Phase 4A run, and an upstream defect of the earlier load (option type defaulting to call) had removed the put rows of the raw table for the dates from about October 2023 to December 2025. The exact inputs were therefore recovered from the surviving Phase 4A batch file (Section 4.1); the inputs of the Phase 3 and 4 baseline runs of November 2025 survive in the same way. Eight dated events fix the lineage:

- 2025-11-20: the Phase 3 (2024) and Phase 4 (2020) baseline batches are run on o4-mini with calendar dates as day labels, 223 windows each (Section 3.6).
- 2026-01-04: Phase 4A batch run (o4-mini-2025-04-16) on 1,476 SPY windows with ends from 2020-02-13 to 2025-12-31.
- 2026-01-15: the pre-rebuild daily table is computed from the same raw table with Black–Scholes gamma and same-day expiries zeroed.
- 2026-02-03: the raw options table is reloaded, with the put rows restored.
- 2026-09-14: the model object gpt-6-luna is created.
- 2026-10-04: the pre-rebuild daily table is archived and treated as read-only.
- 2026-10-08: the daily summary is rebuilt from stored per-contract gamma, same-day expiries included.
- 2026-10-12: equity options move from Alpha Vantage to Cboe delayed quotes, so every window ends on or before 2026-10-09.

Four observations carry the evidence on the put-row loss. First, on each of the 565 trading days from 2023-09-29 to 2025-12-31 the January value equals the call-only GEX of today's raw table, same-day expiries included, to within \$0.005 billion, and it equals the complete net GEX on none of them. Second, through 2023-09-28 the January value equals the complete series on 900 of 940 days; 39 scattered days, the first on 2021-09-17, are also call-only, and one day matches neither. Over whole years the share of days on which the January value equals the complete series is 2020: 100.0%, 2021: 99.2%, 2022: 98.4%, 2023: 60.6%, 2024: 0.0%, 2025: 0.0%. Third, the pre-rebuild table of 2026-01-15 was computed from the same put-less raw table and is therefore also put-less from about October 2023. Fourth, the January calendar lacks 2023-07-12 and 2024-03-11, two trading days that the reloaded table holds; the 58 January windows that span them are flagged (Section 3.4). The upstream defect (the option type defaulted to call) is tracked as issue 24 of the database repository (gex-database-infrastructure).

The pre-rebuild table is therefore not a clean old convention. The effect of Black–Scholes against stored gamma is identifiable only before about October 2023, and the effect of same-day expiries is the difference between `new_full` and `new_ex0dte`.

3.4. Windows and Deterministic Classifier

A window consists of 30 consecutive rows of a series (days $T-29$ to $T+0$), with one window per end date (stride one) and end dates up to 2026-10-09. A span that covers a calendar gap of more

than four days is flagged (a gap of five or more days is not a holiday weekend), and the number of trading days of the symbol's master calendar (`gex.options_daily_summary`) that are absent from the series inside the span is stored with the window; a gap flags a window but does not drop it. Values are relabelled "Day T-29" to "Day T+0" with no dates or tickers, exactly as in Phase 4A, except in the two dated arms of Section 3.6, which carry the ISO dates of the November 2025 prompts. The deterministic classifier of Section 3.1 labels every window independently of any model call, and its verdict is the reference label for each arm. The windows that end in 2026, up to 2026-10-09, form the post-publication period (Section 4.8): the published sample ends on 2025-12-31, so the 29 earliest of these windows still contain days of it, and those with ends from 2026-02-13 lie entirely after it. Table 2 gives the 38,501 windows. The January inputs and the four Phase 3 and 4 inputs keep their own calendars (rule J), and a build stops unless it reproduces the surviving batch file byte for byte. The composite intraday windows (rule T0, key `intraday_t0`) take the `new_full` values for days $T-29$ to $T-1$ and the `final_rush` value at $T+0$. A window with a missing value is dropped. Of the 1,476 January windows, 58 span one of the two missing dates and are flagged. Windows overlap in 29 of 30 days, which the statistics of Section 3.8 take into account.

Table 2. Windows per series and symbol. Rule A cuts 30 consecutive rows of a series that follows the master calendar, rule J does the same for a series with its own calendar, and rule T0 builds the composite intraday windows. The last three columns count windows whose widest calendar gap exceeds four days, windows with at least one master-calendar day absent from the series, and windows with either flag. Flagged windows are kept.

Series key	Symbol	Rule	Windows	Window ends	Gap > 4 d	Missing days	Either
`jan2026_phase4a`	SPY	J	1,476	2020-02-13 to 2025-12-31	0	58	58
`call_only`	SPY	A	1,671	2020-02-13 to 2026-10-09	0	0	0
`old_bs`	SPY	A	4,638	2008-02-13 to 2026-10-02	116	87	174
`new_full`	IWM	A	4,678	2008-02-13 to 2026-10-09	87	0	87
`new_full`	QQQ	A	3,867	2011-05-04 to 2026-10-09	87	0	87
`new_full`	SPY	A	4,671	2008-02-13 to 2026-10-09	87	0	87
`new_ex0dte`	IWM	A	4,678	2008-02-13 to 2026-10-09	87	0	87
`new_ex0dte`	QQQ	A	3,867	2011-05-04 to 2026-10-09	87	0	87
`new_ex0dte`	SPY	A	4,671	2008-02-13 to 2026-10-09	87	0	87
`intraday_t0`	SPY	T0	51	2026-07-27 to 2026-10-09	0	0	0
`new_full_scaled`	IWM	A	1,671	2020-02-13 to 2026-10-09	0	0	0
`new_full_scaled`	QQQ	A	1,670	2020-02-13 to 2026-10-09	0	0	0
`p3_2024`	SPY	J	223	2024-02-13 to 2024-12-31	0	0	0
`p3_2024_dated`	SPY	J	223	2024-02-13 to 2024-12-31	0	0	0
`p4_2020`	SPY	J	223	2020-02-13 to 2020-12-31	0	0	0
`p4_2020_dated`	SPY	J	223	2020-02-13 to 2020-12-31	0	0	0
Total			38,501		638	145	754

3.5. Model Protocol

All model calls use OpenAI's `gpt-6-luna` through the Batch API (`/v1/chat/completions`, 24-hour completion window) in chunks of at most 1,500 requests. The model replaces the retired o4-mini; its API alias carries no dated snapshot, so the creation timestamp of the model object (2026-09-14) and the response identifier of every answer are stored. Requests use `reasoning_effort = medium`, the effort at which o4-mini ran in the published run, and the prompt text is unchanged from the Phase 4A run (prompt version `v2.0_postgresql`); only the two dated arms of Section 3.6 differ in the data lines, whose day labels are ISO dates. The answer has nine fields (Appendix A). The one protocol change that bears on the answer is that these fields are enforced with a strict JSON schema, whereas the earlier runs requested JSON in the prompt text only; the change is disclosed here and does not alter the question asked. The AIAI Methods wording (temperature 1.0, 16,384 tokens) was replaced in the JRFM appendix: the published requests supplied neither a temperature nor a token

limit, and the requests to gpt-6-luna supply no temperature either. They do cap completion at 4,000 tokens, reasoning tokens included, as a safety limit that no answer reached. An answer is read as JSON; a second attempt tolerates raw control characters inside strings (one Phase 4A answer needed it), a last fallback extracts the JSON object from free text, and the parse status of every answer is stored.

A pilot of 40 windows, balanced on the label of the rule (20 detected, 20 not) and run synchronously at standard prices, measured length and cost before the Batch runs. All 40 answers parsed on the first strategy and all 40 verdicts equal the rule and the published o4-mini answer. A request used 1,913 prompt tokens and on average 700 completion tokens, of which 513 were reasoning tokens; the longest answer used 1,063 completion tokens. The pilot cost $0.02 in all. All gpt-6-luna runs, the pilot included, sent 34,712 requests; 34,712 answers parsed (100.0%), 0 needed the fallback parsers and 0 ended with `finish_reason` = `length`, so the retry at 8,000 tokens was never used. A request averaged 1,914 prompt and 734 completion tokens, and the longest answer used 1,864. At Batch prices of $0.05 and $0.25 per million input and output tokens, 1,000 requests cost $0.28. An earlier probe of 2026-10-28 had given about 1,911 prompt and 814 completion tokens per request and $0.30 per 1,000 requests, with 31 of 31 verdicts equal to the rule. In that probe the model undercounted `sign_flips` in 7 of 32 answers. The count of the rule is therefore stored beside the model's own value.

The bridge carries a gate. Cohen's kappa between `bridge_jan` and the 1,412 published o4-mini answers has to be at least about 0.6, the value of the project checklist. The observed kappa is 0.99 (99.6% agreement: 646 windows detected by both models, 5 by gpt-6-luna only, 0 by o4-mini only, 761 by neither), so the gate is passed. A full repeat of `bridge_jan` gives a kappa of 0.98 against the first pass (1,476 pairs), which sets the scale of the run-to-run noise.

3.6. Experimental Arms

Table 3 lists the arms. The arm `r1_o4mini` holds the published o4-mini answers, backfilled from the surviving Phase 4A output file without an API call; 64 of its 1,476 answers were lost to a parsing error in January 2026 and are re-parsed from the stored raw output. Every headline number uses the 1,412 answers of the published result; the 64 re-parsed answers are reported separately (as `r1_o4mini_all`). The arms `r1_p3_o4mini` and `r1_p4_o4mini` are backfilled in the same way from the 223 stored answers of each November 2025 baseline run. The arm `bridge_jan` sends the exact published inputs to gpt-6-luna, which separates the effect of the model from the effect of the inputs. The other arms run gpt-6-luna on the complete series of each symbol; the ex-0DTE arms of QQQ and IWM start in 2020. The intraday arm uses composite windows whose last day is the late-session snapshot.

Two further groups of arms test the earlier results more directly. The Phase 3 (2024) and Phase 4 (2020) baselines of the earlier studies [1,3], the 81.2% versus 12.1% contrast, were run on o4-mini in November 2025 with ISO calendar dates as day labels instead of relative labels. The arms `p34_dated_2024` and `p34_dated_2020` send the same 252 values per year, with the dates that were actually sent, to gpt-6-luna; the arms `p34_obf_2024` and `p34_obf_2020` send the identical values with Day-T labels. Within a year the two arms differ in the labels only, which isolates the effect of visible dates, and each is compared with the stored o4-mini answers. The two baselines are not on one convention (Table 1): the 2024 series is a gross call-plus-put magnitude with a negative sign imposed and the 2020 series is a magnitude without sign. We keep them verbatim because the aim is to replay the published contrast, which therefore cannot be read as one between a negative and a positive regime.

With the thresholds fixed in dollars, QQQ and IWM have no persistent window (Section 4.7), so the unscaled arms only test whether the model also rejects every window. As a disclosed protocol change, the arms `full_qqq_scaled` and `full_iwm_scaled` therefore run on `new_full_scaled`: the `new_full` values of QQQ and IWM multiplied by one fixed factor per symbol, the median over 2020-01-01 to 2026-10-09 of SPY's open-interest notional divided by the symbol's (Appendix B). The factors are 4.22 for QQQ (over 1,699 days) and 11.43 for IWM (over 1,700 days), and they are stored

with the series. Prompt, thresholds and classifier are unchanged, so scaled and unscaled arms are labelled by the same rule and run side by side.

Table 3. Experimental arms and run matrix. Windows, requests and cost come from the run ledger (Table A4). Requests include the test-retest repeats, and the pilot ran synchronously at standard prices. Batch costs use $0.05 and $0.25 per million input and output tokens. The three o4-mini backfills are not spend of this study.

Arm	Series	Symbol	Window ends	Windows	Requests	Cost (USD)
r1_o4mini	jan2026_phase4a	SPY	2020–2025	1,476	0 (backfill)	–
r1_p3_o4mini	p3_2024_dated	SPY	2024	223	0 (backfill)	–
r1_p4_o4mini	p4_2020_dated	SPY	2020	223	0 (backfill)	–
bridge_jan	jan2026_phase4a	SPY	2020–2025	1,476	2,952	0.82
full_spy	new_full	SPY	2008–2026	4,671	4,905	1.44
ex0dte_spy	new_ex0dte	SPY	2008–2026	4,671	4,905	1.40
p34_obf_2024	p3_2024	SPY	2024	223	234	0.06
p34_dated_2024	p3_2024_dated	SPY	2024	223	234	0.06
p34_obf_2020	p4_2020	SPY	2020	223	234	0.06
p34_dated_2020	p4_2020_dated	SPY	2020	223	234	0.06
full_qqq	new_full	QQQ	2011–2026	3,867	4,060	1.16
full_iwm	new_full	IWM	2008–2026	4,678	4,912	1.26
full_qqq_scaled	new_full_scaled	QQQ	2020–2026	1,670	1,754	0.53
full_iwm_scaled	new_full_scaled	IWM	2020–2026	1,671	1,755	0.50
ex0dte_qqq	new_ex0dte	QQQ	2020–2026	1,699	1,784	0.50
ex0dte_iwm	new_ex0dte	IWM	2020–2026	1,700	1,785	0.46
intraday_spy	intraday_late	SPY	2026-07 to 2026-10	51	54	0.02
bridge_old	old_bs	SPY	2008–2026	4,638	4,870	1.36
pilot_bridge_jan	jan2026_phase4a	SPY	2020–2025	40	40 (sync)	0.02
Total					34,712	9.70

The runs went in priority order. The first group was `bridge_jan`, `full_spy`, `ex0dte_spy` and the four `p34_*` arms; the second the unscaled and scaled arms of QQQ and IWM; the third the ex-0DTE arms of QQQ and IWM and the intraday arm; `bridge_old` was optional and ran last. Test-retest duplicates carry the suffix `#rep1`: `bridge_jan` is repeated in full (1,476 requests), and every other arm carries a duplicate sample of about 5% of its windows (1,512 requests in all). The first pass alone feeds every rate, and the duplicates feed the agreement of repeated runs. A request that ends with `finish_reason` = `length` is repeated once with 8,000 tokens; none was needed. All new requests cost $9.70 in total, against a hard stop of $12.00. Only runs paid in this study count: the three o4-mini backfills, worth $9.97 at o4-mini Batch prices, do not. The scaling of QQQ and IWM was decided on 2026-10-28, and both variants are run side by side. It is a protocol change that is disclosed as such, not a re-tuning of the thresholds.

3.7. Storage and Reproducibility

Every series value, window, classifier verdict, run and model answer is written to four tables, `gex_llm_regime_series`, `gex_llm_regime_windows`, `gex_llm_regime_runs` and `gex_llm_regime_detections`, with a summary view of detection rates per arm and year. The tables exist as parquet files under `data/backups/gex_llm/tables/`, and, after the documentation pull request in the database repository was approved, as standalone tables without foreign keys into the core tables in the `backups` schema of the project database; the schema was applied and the parquet copies were loaded on 2026-10-30. The raw batch input and output files, the batch object and a parquet copy of the detections are kept per run under `data/backups/gex_llm/<run_id>/`, with run identifiers of the form `<arm>_<model>_<mode>_<UTC timestamp>`; the three backfilled o4-mini runs are `r1_o4mini_phase4a_20260104`, `r1_p3_o4mini_20251120` and `r1_p4_o4mini_20251120`, and a test-retest repeat is stored under the arm `<arm>#rep1`. All tables and figures of this article are

generated from the stored tables and from the stored Phase 3, 4 and 4A files; no figure falls back to synthetic data. Every window stores the SHA-256 of its input, every run row the hashes of its batch input and output files, and `report.yaml` the hashes of all result files and a fingerprint of the stored tables (beginning 3b69bd38197d). The project keeps three copies of every result: the database tables, the parquet files and the raw batch files on the local disk. Nothing raw is in the repository or on its remote. The raw files and the full tables, which hold the model text, are ignored by git; the repository keeps the run provenance, the condensed per-window tables (detections without text, windows without sequences, a series index) and the result files of this article. Every number of the article can be reproduced from the condensed tables. The run ledger is Table A4 in Appendix C.

3.8. Statistics

Detection rates are reported with 95% Wilson intervals [21]. Windows have stride one, so neighbouring windows share 29 of their 30 days and cannot be treated as independent. Every p-value is therefore accompanied by (i) a calendar-quarter block bootstrap with 2,000 draws (seed 261) and (ii) the same statistic on a non-overlapping 30-day subsample, the greedy chain of windows that share no day, starting at the first window of the period. Agreement between models is measured with Cohen's kappa, and paired detection rates across series or arms with the exact McNemar test (on the subsample; the stride-one value is reported as anti-conservative); effect sizes between years use φ. All p-values of the AIAI and JRFM papers [1,3] treated the windows as independent. The quarter-block bootstrap resamples the calendar quarters of the window end with replacement, and a paired contrast resamples the same quarters for both series. Its two-sided p-value is $2\min\{P(d \leq 0), P(d \geq 0)\}$, bounded below by $1/2{,}001$, and an interval built from fewer than about eight quarters is coarse. A single year gives four quarters, and the period 2024 to 2025 gives 8. The non-overlapping subsample depends on its first window, so the 30 possible first windows give 30 chains, and the range of their rates is reported beside the rate of the chain that starts at the first window. For the rule on `new_full` (SPY, window ends in 2024 and 2025), the chain holds 3 of 17 windows (17.6%), and the rate ranges from 5.9% to 25.0% across the starts. Between periods, φ is signed (late minus early) and the p-value comes from Fisher's exact test. Each p-value in Section 4 names the method it uses.

4. Results

4.1. What the Published Inputs Were

Table 4 compares the inputs recovered from the Phase 4A batch file with each candidate series, and Figures 2 and 3 show one window and the 2008–2026 history. The batch file of 2026-01-04 stores the daily values that were sent to o4-mini. They give 1,476 windows with ends from 2020-02-13 to 2025-12-31, cut from 1,505 daily values. Compared with the complete series, the January calendar lacks 2023-07-12 and 2024-03-11. The 58 windows that span one of the two gaps are flagged and kept (Section 3.4).

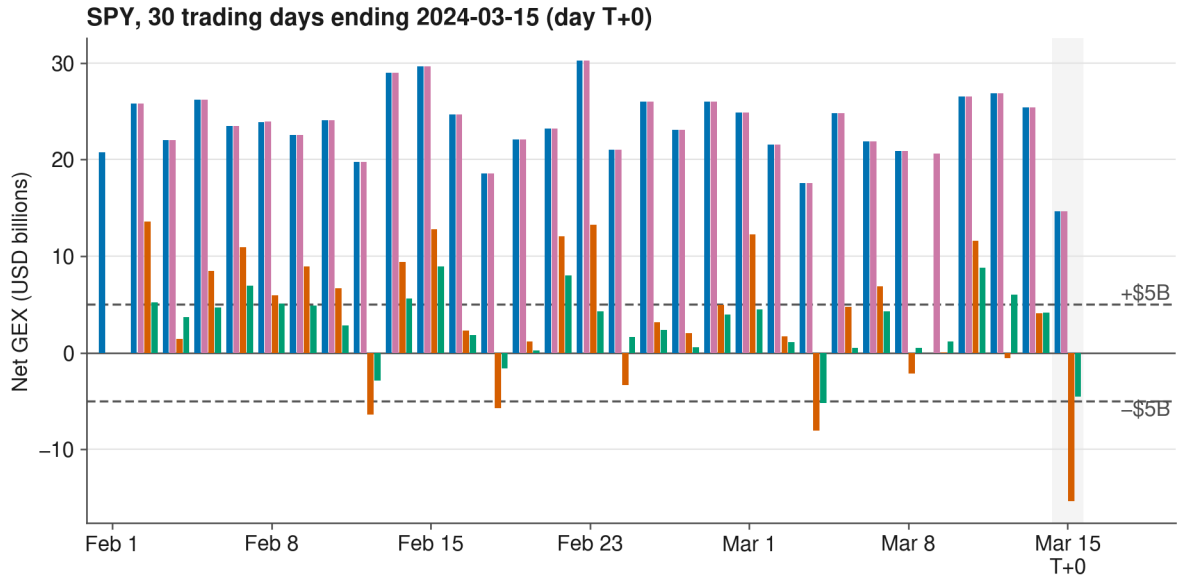
The published inputs follow two different series (Table 4). In 2020 to 2022 the share of daily values that equal `new_full` at the 0.01B precision of the prompt is 100.0%, 99.2% and 98.4%, and the call-only series equals the inputs on at most 1.2% of the days. In 2024 and 2025 the relation is reversed: none of the values equals `new_full` (0.0% in both years) and every value equals the call-only GEX (100.0% in both years). The year 2023 is the transition: 60.6% of its values equal `new_full` and 39.4% equal the call-only series, which together account for every day. From 2023-09-29 on, every value equals the call-only GEX; the day before, 2023-09-28, is the last on which a value differs from it. Before that date, 39 days (the first on 2021-09-17) already equal the call-only value and not the complete one. The published inputs of 2024 and 2025 are therefore the call leg of the options chain without the put leg, and they differ in level and sign from the complete series: the mean |GEX| of the published inputs is $20.19 billion in 2024 against $6.48 billion on the complete series, and $19.21 billion against $7.13 billion in 2025, with the same daily sign on 62.2% and 54.0% of the days.

The pre-rebuild archive (`old_bs`) was computed from the same put-less table for the same dates, so it cannot serve as a clean earlier convention. Its Black–Scholes gamma differs from the stored gamma, and none of its values equals a published value at 0.01B (0.0% in every year). Its signs agree with the published inputs on 100.0% of the days in 2024 and 99.6% in 2025, and its mean |GEX| is \$16.19 billion and \$14.71 billion, which is the level of a call-only series. The archive also has holes: it lacks 33 days of the complete calendar (2023-07-12, 2024-03-11, the 26 days from 2026-01-02 to 2026-02-09, and the 5 days from 2026-10-05 to 2026-10-09).

The inputs of the Phase 3 and 4 runs of November 2025 survive in the same way: 252 daily values for 2024 and 252 for 2020, 223 windows each, with ISO dates in the prompts. The 2024 values are a gross call-plus-put magnitude with a negative sign imposed. They are negative on 242 days, and the 10 positive values (\$10.61 billion to \$13.76 billion) are placeholder rows of the source database. The 2020 values are a magnitude only and are positive on every day. The mean absolute daily value is \$31.24 billion for 2024 and \$2.85 billion for 2020, so the two baselines differ in convention and in scale (Section 4.11).

Table 4. Agreement of the recovered published inputs with each candidate series, by year (share of daily values equal at the 0.01B precision of the prompt, in percent). The archive uses another gamma model, so exact equality is not expected for it.

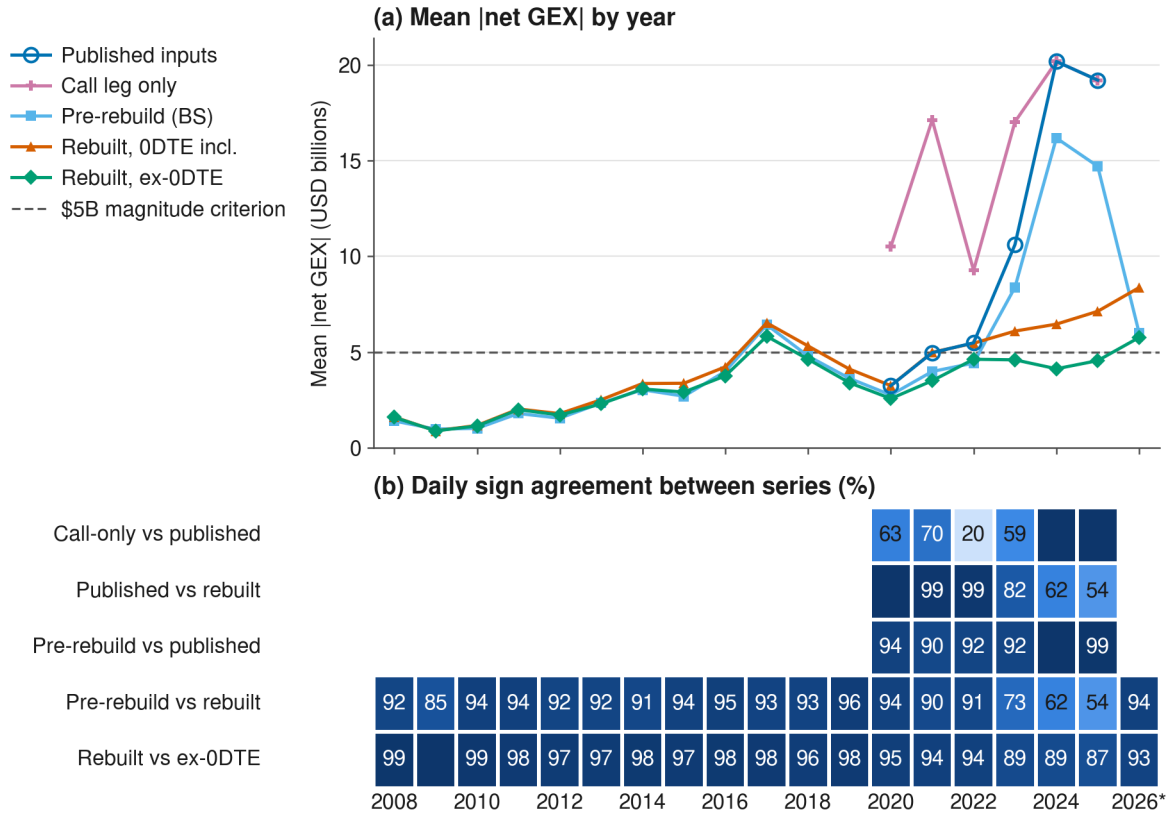
Year	Days	`new_full`	`call_only`	`old_bs`
2020	252	100.0	0.0	0.0
2021	252	99.2	0.8	0.0
2022	251	98.4	1.2	0.0
2023	249	60.6	39.4	0.0
2024	251	0.0	100.0	0.0
2025	250	0.0	100.0	0.0



Series, in bar order	Mean |GEX| (USD billions)	Days positive	Sign flips	Rule verdict
Published inputs	23.6	30 of 30	0	Detected, positive
Call leg only	23.6	30 of 30	0	Detected, positive
Rebuilt, 0DTE incl.	6.7	23 of 30	13	Not detected, low conviction
Rebuilt, ex-0DTE	3.9	26 of 30	7	Not detected, low conviction

Dashed lines: the $5B magnitude criterion of the classifier (mean |GEX| of the 30 days). Published inputs: no row for 2024-03-11, which the other series have, so its window starts on 2024-02-01.

Figure 2. One SPY 30-day window under four conventions (`jan2026_phase4a`, `call_only`, `new_ex0dte`, `new_full`), window ending 2024-03-15. Dashed lines mark the ±$5 billion magnitude threshold; the table under the bars gives the three criteria and the verdict for each convention. **Read this figure as:** the same 30 trading days receive different verdicts. The published inputs and the call-only series have 30 of 30 positive days, a mean |GEX| of $23.56 billion, 0 sign flips, and are detected. The complete series has 23 positive days, 13 flips and a mean of $6.66 billion, and the ex-0DTE series 26 positive days, 7 flips and $3.87 billion; both are rejected as low conviction. On the last day the net GEX is +$14.61 billion on the published inputs, −$15.35 billion on the complete series and −$4.51 billion without same-day expiries.



SPY, daily values. (a) Yearly mean over the days a series shares with its comparison partner; the published inputs are drawn hollow, on top. (b) Share of days on which both values are positive or both are not; dark cells without a number are 100%, blank cells have no days that year. * 2026: trading days up to 2026-10-09.

Figure 3. SPY, 2008–2026. (a) Yearly mean of |GEX| under each series, with the $5 billion magnitude criterion for scale. (b) Daily sign agreement between pairs of series, by year. **Read this figure as:** until 2022 the published inputs and the complete series are close; from 2023 the published inputs rise to $20.19 billion in 2024 while the complete series stays between $6.05 billion and $7.13 billion in 2023 to 2025. Their daily sign agreement falls from 98.8% in 2022 to 81.5%, 62.2% and 54.0%. The call-only series matches the published inputs in 2024 and 2025 only.

4.2. Classifier Replay by Year

Table 5 and Figure 4 give the detection rate of the deterministic classifier by year for each series next to the published o4-mini rates, and Figure 5 shows how far the replay depends on the thresholds. On the recovered inputs the rule reproduces the published labels in 1,412 of 1,412 parsed windows (100.0%; Cohen's kappa 1.000), so the published labels follow from the published inputs and the published criteria. On the published inputs the rule detects 251 of 251 (100.0%) of the windows ending in 2024 and 250 of 250 (100.0%) in 2025, and the published o4-mini answers give 241 of 241 (100.0%) and 245 of 245 (100.0%) (Wilson 95% lower bound 98.4% in 2024 and 98.5% in 2025). The call-only series and the archive give the same saturation (100.0% in both years on the call-only series and 100.0% on the archive). On the complete series the rule detects 26 of 252 (10.3%) in 2024 (Wilson 95% interval 7.1–14.7%; quarter-block bootstrap 3.2–17.3%, 4 blocks) and 34 of 250 (13.6%) in 2025 (Wilson 95% interval 9.9–18.4%; bootstrap 2.3–25.4%). Without same-day expiries the rates are 13.1% and 16.4%.

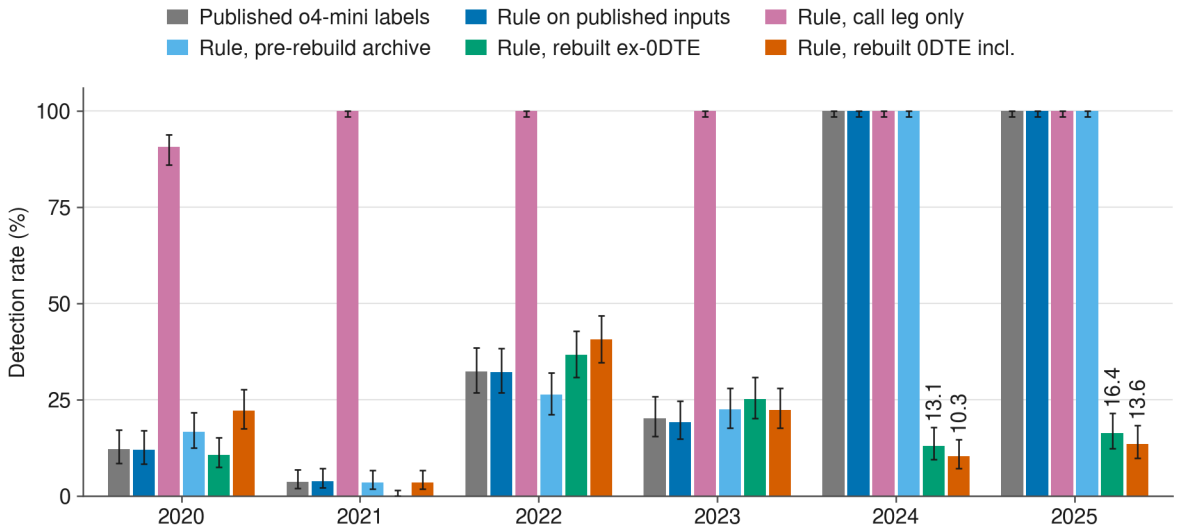
Pooled over 2024 and 2025 the detection rate is 100.0% on the published inputs (501 of 501 windows) and 12.0% on the complete series (60 of 502; Wilson 9.4–15.1%, quarter-block bootstrap 5.0–19.2%; non-overlapping subsample 3 of 17, and between 5.9% and 25.0% over the 30 possible starting windows). The paired difference is 88.0 percentage points (quarter-block bootstrap interval 80.8–95.0, $p = 5.0 \times 10^{-4}$, the smallest value that 2,000 draws can give; exact McNemar test on the non-overlapping subsample $p = 1.2 \times 10^{-4}$).

The period contrast changes sign. On the published inputs the rule detects 17.0% of the windows of 2020 to 2023 and 100.0% of those of 2024 and 2025, a shift of 83.0 percentage points ($\varphi = 0.789$; bootstrap $p = 5.0 \times 10^{-4}$; Fisher exact test on the non-overlapping subsample $p = 2.7 \times 10^{-9}$). On the complete series the rates are 22.2% and 12.0%, a shift of -10.2 percentage points ($\varphi = -0.124$; bootstrap $p = 0.108$; Fisher exact test on the subsample $p = 0.728$): the complete series shows no increase after 2023 that the data distinguish from zero. In 2020 to 2023 the two series are of the same order (Table 5).

Across the 45 threshold sets the rate on the published inputs stays at 100.0%, whereas the rate on the complete series runs from 4.0% to 29.3%. The gap between the two is at least 70.7 and at most 96.0 percentage points in every configuration, so the difference between the published and the complete inputs does not depend on the exact thresholds.

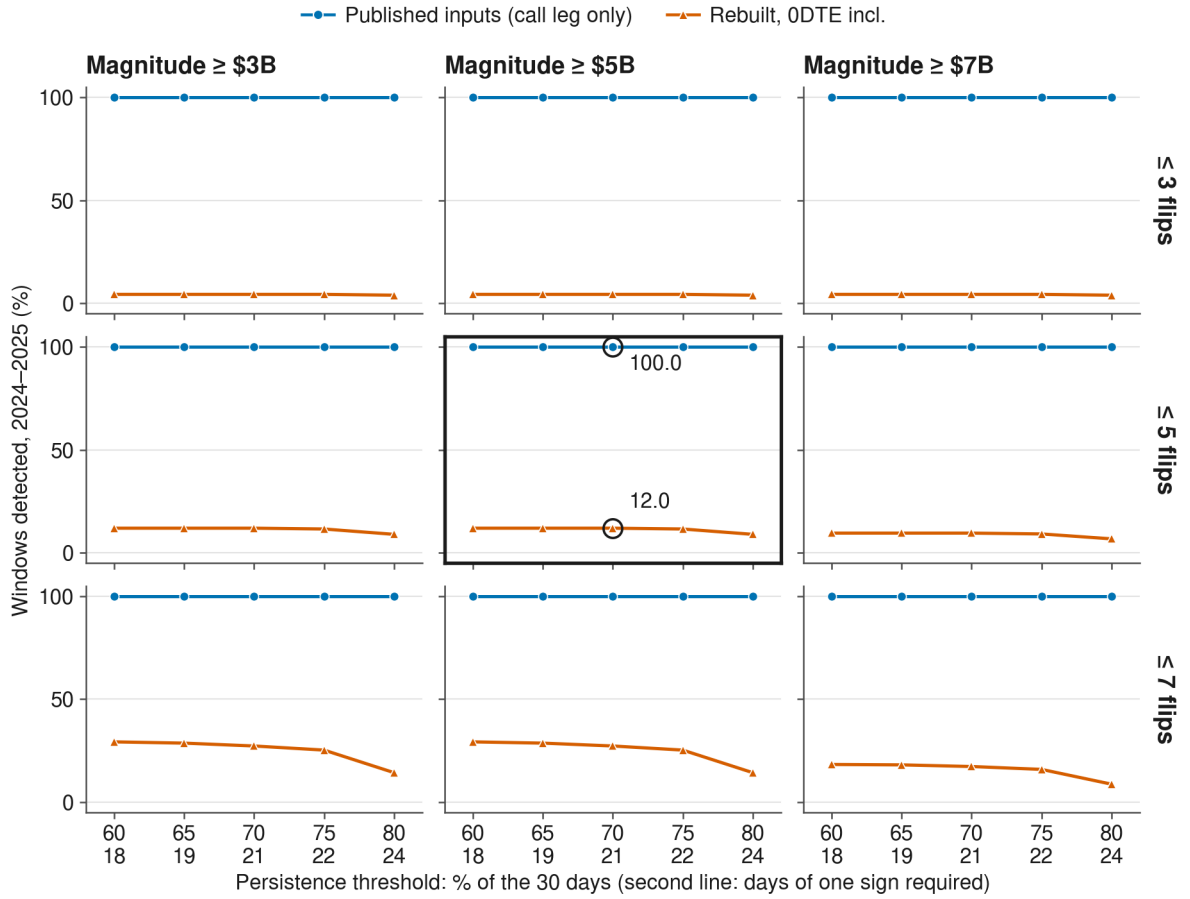
Table 5. Detection rate of the deterministic classifier by year, SPY, in percent (windows detected out of windows, below the rate), next to the published o4-mini rate. The Windows column gives the windows of `new_full`; every cell shows its own count. The published rate covers the 1,412 answers of the published result.

Year	Windows	Published	`jan2026_phase4a`	`call_only`	`old_bs`	`new_ex0dte`	`new_full`
		12.2	12.1	90.6	16.7	10.7	22.2
2020	252	26/213	27/223	202/223	42/252	27/252	56/252
		3.7	4.0	100.0	3.6	0.0	3.6
2021	252	9/241	10/252	252/252	9/252	0/252	9/252
		32.4	32.3	100.0	26.3	36.7	40.6
2022	251	79/244	81/251	251/251	66/251	92/251	102/251
		20.2	19.3	100.0	22.5	25.2	22.4
2023	250	46/228	48/249	250/250	56/249	63/250	56/250
		100.0	100.0	100.0	100.0	13.1	10.3
2024	252	241/241	251/251	252/252	251/251	33/252	26/252
		100.0	100.0	100.0	100.0	16.4	13.6
2025	250	245/245	250/250	250/250	250/250	41/250	34/250



SPY, 30-day windows by year of the window end. Error bars: Wilson 95% CI (overlapping windows treated as independent, so optimistic); a year holds only 8–9 non-overlapping windows. First bar: o4-mini output of January 2026 (parsed answers for 1,412 of 1,476 windows; the 64 answers re-parsed in 2026 are not drawn); other bars: the rule applied to all windows of the series in the legend.

Figure 4. Detection rate by year, SPY, 2020–2025: the published o4-mini labels and the deterministic rule replayed on the published inputs (`jan2026_phase4a`), the call leg only (`call_only`), the pre-rebuild archive (`old_bs`) and the rebuilt table without and with same-day expiries (`new_ex0dte`, `new_full`), with 95% Wilson intervals. **Read this figure as:** the published labels and the rule on the published inputs agree window by window. In 2024 and 2025 the bars of the published inputs, the call-only series and the archive stand at 100%, and the two bars of the rebuilt series stand at 13.1% and 10.3% in 2024 and 16.4% and 13.6% in 2025. The Wilson intervals are optimistic because the overlapping windows of a year are treated as independent.



SPY windows ending in 2024 or 2025 (501 on the published inputs, 502 on the rebuilt series), each re-classified from its stored 30-day sequence. Heavy frame and rings: the published thresholds (70%, $5B, 5 flips). Across the 45 threshold sets the published-input rate exceeds the rebuilt rate by 70.7 to 96.0 percentage points.

Figure 5. Sensitivity of the 2024–2025 detection rate to the persistence, magnitude and flip thresholds (45 threshold sets: one panel per magnitude and flip limit, persistence on the horizontal axis) for the published inputs (`jan2026_phase4a`) and the rebuilt series (`new_full`); the panel with the heavy frame holds the published thresholds. **Read this figure as:** the curve of the published inputs is flat at 100.0% in all nine panels, and the curve of the rebuilt series stays low and moves with the flip limit: it is highest, up to 29.3%, with a limit of seven flips and lowest, 4.0%, with three. At the published thresholds the rebuilt series gives 12.0%.

4.3. Model Agreement on the Published Inputs

Table 6 and Figure 6 compare gpt-6-luna with the published o4-mini answers on identical inputs. The agreement gate of Section 3.5 (Cohen's kappa of at least 0.6) is met with a wide margin. On the 1,412 windows with a published answer the two models give the same verdict in 1,407 windows (99.6%; quarter-block bootstrap interval 99.4–99.9%), and kappa is 0.993 (bootstrap interval 0.986–0.998). The confusion matrix has 646 windows detected by both, 761 by neither, 0 only by o4-mini and 5 only by gpt-6-luna. In each of the 5 windows of disagreement the deterministic rule agrees with o4-mini. The windows lie in 2021 to 2023 (3 in 2021, 1 in 2022, 1 in 2023); the yearly kappa is lowest in 2021 (0.851) and between 0.986 and 1.000 in 2020, 2022 and 2023, and it is undefined in 2024 and 2025, where both models label every window as detected. The detection rates are 45.8% for o4-mini and 46.1% for gpt-6-luna on these windows. The paired McNemar test on all windows at stride one gives $p = 0.063$ (anti-conservative, because the windows overlap) and the quarter-block bootstrap $p = 0.010$ (Table 6 gives both). No McNemar test on the non-overlapping subsample was run for this comparison.

The 64 answers of the published run that were lost to a parsing error in January 2026 and re-parsed from the stored raw output are held out of every headline number. With them the comparison covers

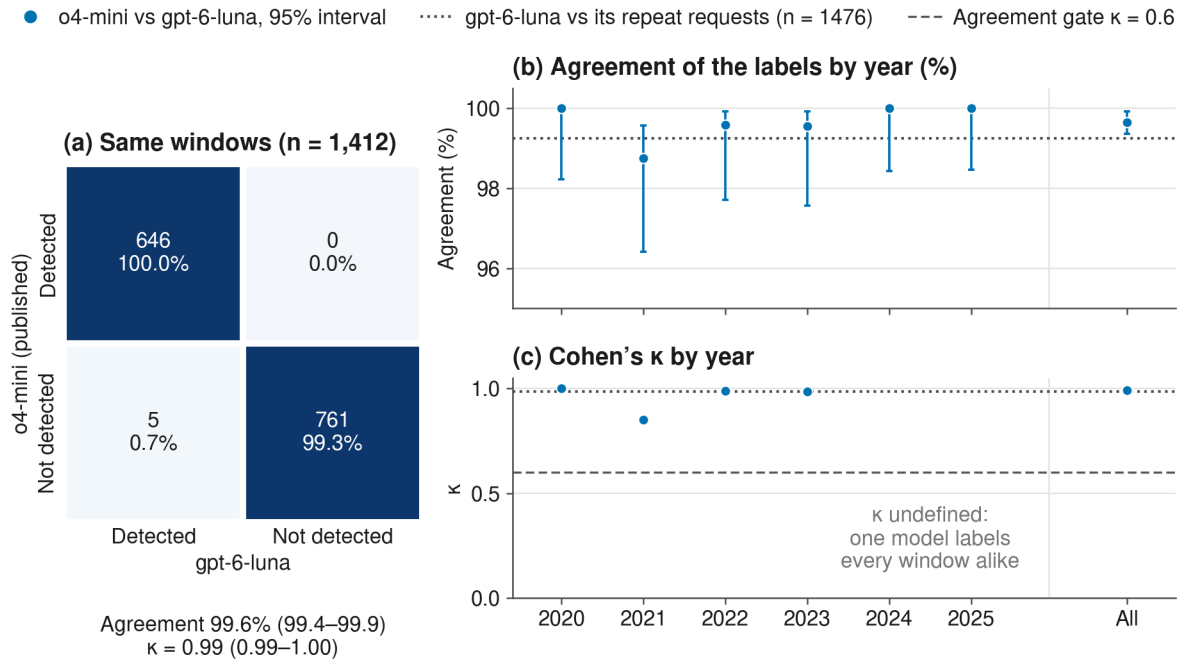
1,476 windows: agreement 99.6%, kappa 0.992 (bootstrap 0.984–0.997), and 6 windows of disagreement, all with the rule on the side of o4-mini. The conclusions do not depend on whether they are included.

On the exact published inputs the mean magnitude that o4-mini reports equals the mean computed from the window: the median ratio of reported to computed is 1.000, and 99.9% of the answers lie within 5% of the computed value. For gpt-6-luna the median ratio is 1.000 (99.7% within 5%). The pre-rebuild archive cannot serve for this check: its daily values are 0.818 times the published inputs at the median (ratio of the archive to the published inputs), so a comparison with the archive would suggest a magnitude error that the exact inputs rule out.

The repeat pass runs gpt-6-luna a second time on all 1,476 windows of the published inputs. The two passes agree on 99.3% of the windows (kappa 0.985, bootstrap 0.969–0.996) and differ in 11 windows, which is more than the 5 windows in which the first pass differs from o4-mini. The disagreement between the two models is thus no larger than the disagreement of the model with itself. The repeats of the other arms agree on 98.8% to 100.0% of their windows (Table A4 lists the runs; the repeat of `full_spy` agrees on 99.6% of 234 windows).

Table 6. Bridge to the published result: gpt-6-luna versus o4-mini on the exact Phase 4A inputs. The first block holds the 1,412 answers of the published result, the second adds the 64 re-parsed answers, and the third compares gpt-6-luna with its own repeat pass. Intervals are quarter-block bootstrap intervals; the McNemar test is exact on all windows at stride one and therefore optimistic.

Statistic	o4-mini	gpt-6-luna	Both
Windows with a parsed answer	1,412	1,412	1,412
Detection rate (%)	45.8	46.1	–
Agreement with the classifier (%)	100.0	99.6	–
Agreement between models (%) [95% block-bootstrap interval]	–	–	99.6 [99.4, 99.9]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.993 [0.986, 0.998]
McNemar exact p (stride one, all windows)	–	–	0.063
Block-bootstrap p (calendar quarters)	–	–	0.010
Median reported / computed mean |GEX|	1.000	1.000	–
With the 64 re-parsed answers			
Windows with a parsed answer	1,476	1,476	1,476
Detection rate (%)	45.2	45.5	–
Agreement with the classifier (%)	100.0	99.6	–
Agreement between models (%) [95% block-bootstrap interval]	–	–	99.6 [99.3, 99.9]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.992 [0.984, 0.997]
McNemar exact p (stride one, all windows)	–	–	0.219
Block-bootstrap p (calendar quarters)	–	–	0.121
Test-retest of gpt-6-luna (first pass against repeat pass)			
Windows with a parsed answer	–	1,476	–
Agreement between passes (%) [95% block-bootstrap interval]	–	–	99.3 [98.6, 99.8]
Cohen's kappa [95% block-bootstrap interval]	–	–	0.985 [0.969, 0.996]



SPY windows of the recovered Phase 4A inputs, window ends 2020–2025. Whiskers on years: Wilson 95% intervals, which treat overlapping windows as independent; All: quarter-block bootstrap 95% interval. In (a) the percentage is the share of the row.

Figure 6. Bridge to the published result on the Phase 4A windows. (a) Confusion matrix of gpt-6-luna against the o4-mini labels, with the overall agreement and Cohen's kappa. (b), (c) Agreement and kappa by year, with the test-retest of gpt-6-luna and the agreement gate as dashed references. **Read this figure as:** nearly all windows lie on the diagonal (646 and 761), and the five off-diagonal windows are all detections by gpt-6-luna alone. Yearly agreement is 98.8% or higher and the yearly kappa is far above the gate of 0.6, at 0.851 in its lowest year. The test-retest agreement of gpt-6-luna with itself (99.3%) is a lower reference than the agreement with o4-mini.

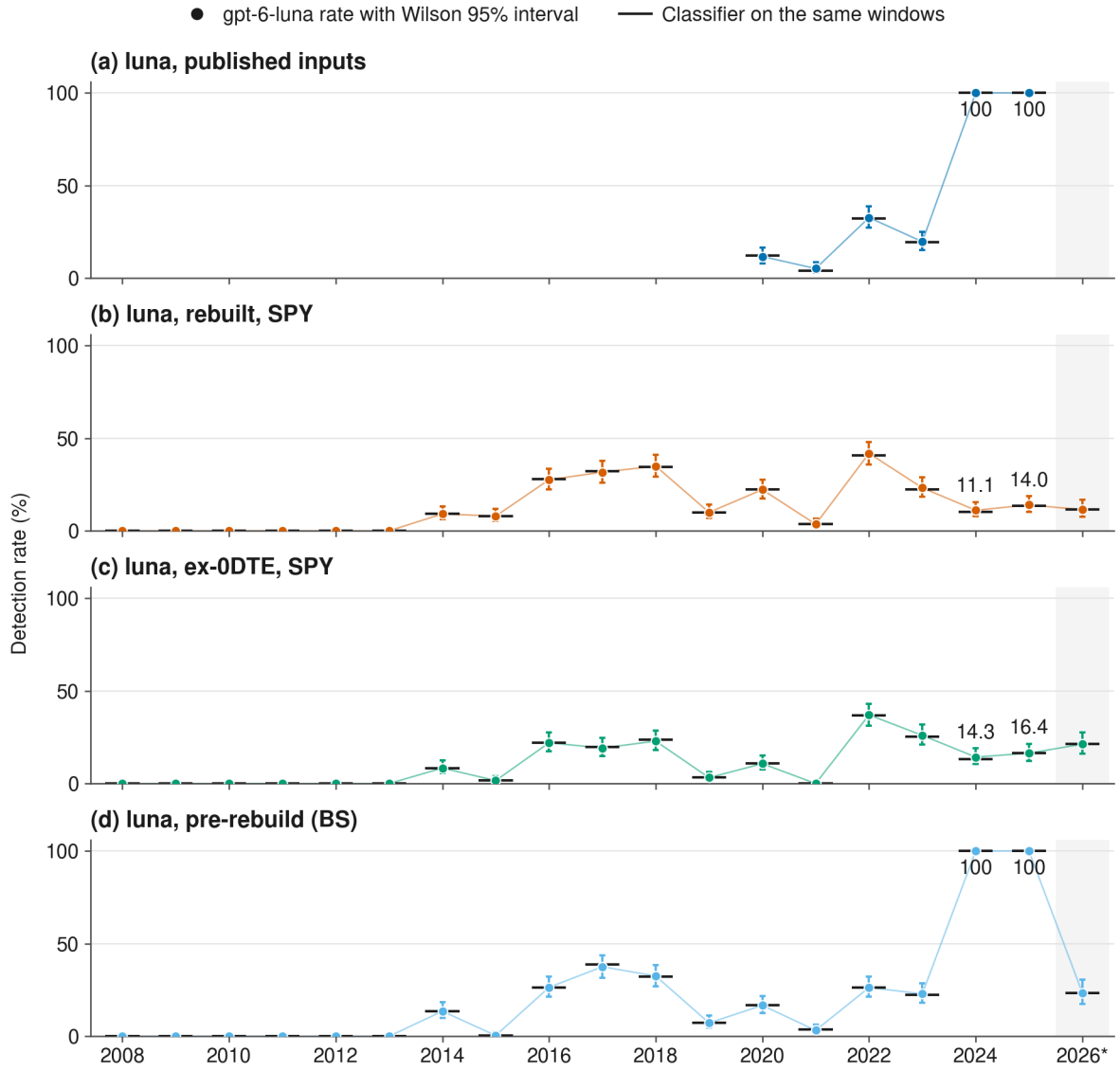
4.4. gpt-6-luna on the Complete Series

Figure 7 shows the detection rate of gpt-6-luna by year on each SPY series, from the published inputs to the complete series, next to the classifier, and Table 7 gives the yearly rates of the two complete series with their intervals. On the complete series gpt-6-luna labels 613 of 4,671 windows as persistent (13.1%; 606 of them by the rule, 13.0%) and, without same-day expiries, 494 of 4,671 (10.6%; rule 491, 10.5%). In 2024 and 2025 it detects 28 of 252 (11.1%) and 35 of 250 (14.0%) of the windows of the complete series (Wilson 7.8–15.6% and 10.2–18.8%) and 14.3% and 16.4% of those without same-day expiries. Pooled over the two years the rate is 12.5% (63 of 502; Wilson 9.9–15.7%, quarter-block bootstrap 6.1–19.6%, non-overlapping subsample 3 of 17). The same model on the published inputs detects 100.0% of the windows of these two years (501 of 501), and on the pre-rebuild archive 100.0% (501 of 501). The paired difference between the published inputs and the complete series is 87.4 percentage points (quarter-block bootstrap interval 80.3–93.9, $p = 5.0 \times 10^{-4}$; exact McNemar test on the non-overlapping subsample $p = 1.2 \times 10^{-4}$), and that between the archive and the complete series is 87.4 percentage points ($p = 5.0 \times 10^{-4}$ and $p = 1.2 \times 10^{-4}$). In 2020 to 2023 the model detects 22.7% of the windows of the complete series. In 2026 it detects 11.4% of the windows of the complete series and 21.2% of those without same-day expiries (Section 4.8).

The model follows the rule on the complete series. The verdicts agree in 99.7% of the 4,671 windows of `full_spy` (kappa 0.986) and in 99.7% of those of `ex0dte_spy` (kappa 0.985). The two arms disagree with the rule in 15 and 13 windows; the causes are analysed in Section 4.10, and `results/disagreements.csv` lists every window. All 34,712 stored gpt-6-luna answers were parsed at the first attempt: every answer ended with finish reason `stop`, and none needed a fallback parse or the retry at the larger token limit.

Table 7. gpt-6-luna on the complete SPY series, by year of the window end: windows, detection rate in percent with its 95% Wilson interval (overlapping windows treated as independent, hence optimistic) for `full_spy` and `ex0dte_spy`, and the agreement of the model verdict with the rule on the same windows (percent). * Windows ending in 2026 up to 2026-10-09.

		`full_spy`		`ex0dte_spy`		Agreement with rule	
Year	Windows	Rate	Wilson 95%	Rate	Wilson 95%	full	ex-0DTE
2008	224	0.0	0.0–1.7	0.0	0.0–1.7	100.0	100.0
2009	250	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2010	251	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2011	250	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2012	250	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2013	252	0.0	0.0–1.5	0.0	0.0–1.5	100.0	100.0
2014	252	9.1	6.2–13.3	8.3	5.5–12.4	100.0	100.0
2015	252	7.9	5.2–11.9	1.6	0.6–4.0	100.0	100.0
2016	250	27.6	22.4–33.4	22.0	17.3–27.5	99.6	100.0
2017	240	31.7	26.1–37.8	19.2	14.7–24.6	99.6	99.6
2018	249	34.9	29.3–41.0	22.9	18.1–28.5	99.6	99.2
2019	251	10.0	6.8–14.3	3.2	1.6–6.2	100.0	100.0
2020	252	22.2	17.5–27.8	10.7	7.5–15.1	100.0	100.0
2021	252	3.6	1.9–6.6	0.0	0.0–1.5	99.2	100.0
2022	251	41.8	35.9–48.0	37.1	31.3–43.2	98.8	99.6
2023	250	23.2	18.4–28.8	26.0	21.0–31.8	99.2	98.4
2024	252	11.1	7.8–15.6	14.3	10.5–19.1	98.4	98.8
2025	250	14.0	10.2–18.8	16.4	12.3–21.5	99.6	99.2
2026*	193	11.4	7.6–16.7	21.2	16.1–27.5	100.0	100.0



SPY, 30-day windows ending in the year. Whiskers are Wilson intervals that treat the overlapping windows of a year as independent, so they are too narrow: a year holds about 9 non-overlapping 30-day windows. Grey band: 2026, the post-publication period (windows ending up to 2026-10-09).

Figure 7. gpt-6-luna detection rate by year, SPY: one panel per arm, from the published inputs (`bridge_jan`) to the rebuilt series with and without same-day expiries (`full_spy`, `ex0dte_spy`) and the pre-rebuild archive (`bridge_old`). Dots are the model with 95% Wilson intervals, dashes the classifier on the windows the model answered, and the grey band is 2026. **Read this figure as:** on the published inputs (a) and on the archive (d) the rate jumps to 100% in 2024 and 2025. On the two rebuilt series (b, c) it stays between 3.6% and 41.8% (`full_spy`) and between 0.0% and 37.1% (`ex0dte_spy`) from 2020 to 2026, with the highest value in 2022. In every panel the dots lie close to the dashes of the classifier.

4.5. 0DTE Ablation

Table 8 separates the contribution of same-day expiries by comparing `new_full` with `new_ex0dte`. The share of same-day expiries in the absolute net GEX of SPY grows from 0.9% in 2008 and 11.5% in 2013 to 22.1% in 2019, 25.7% in 2020 and 45.9% in 2024, and it stands at 43.9% in 2025 and 40.8% in 2026 (the share is the sum of the absolute same-day-expiry contribution over the sum of the absolute net GEX of `new_full` over the days of the year). For QQQ it is 45.5% in 2025 and for IWM 23.0%. The sign of the daily net GEX changes when same-day expiries are removed on 11.1% of the days of 2024 and 13.2% of the days of 2025 (4.4% of all days).

The effect on the verdict is small and has no constant direction. In 2024 the rule detects 10.3% of the windows of the complete series and 13.1% of those without same-day expiries; 9 windows are detected only on the complete series and 16 only without same-day expiries (difference -2.8 percentage points; quarter-block bootstrap interval $[-10.7, 5.5]$, $p = 0.499$; exact McNemar test on the non-overlapping subsample $p = 1.000$). The counts for 2025 are 4 and 11 (13.6% against 16.4%). Pooled over 2024 and 2025 the rates are 12.0% and 14.7% for the rule (difference -2.8 percentage points, interval $[-8.8, 2.8]$, $p = 0.369$; McNemar on the subsample of 17 windows $p = 1.000$) and 12.5% and 15.3% for gpt-6-luna (-2.8 percentage points, $p = 0.376$; subsample $p = 1.000$). Removing same-day expiries therefore does not restore the high rates of the published inputs: the rate is 14.7% without them and 12.0% with them, against 100.0% on the published inputs. In 2020 to 2023 the complete series detects more than the ex-0DTE series (22.2% against 18.1%, 4.1 percentage points; $p = 0.137$ by bootstrap, $p = 1.000$ by McNemar on the subsample), and in 2026 it detects less (11.4% against 21.2%; 0 windows against 19 in the paired comparison; bootstrap $p = 0.646$, McNemar on the subsample $p = 1.000$). For QQQ and IWM the rule detects no window under either series, so the ablation has no effect on their verdicts.

The window of Figure 2 gives an example. The net GEX of SPY on 2024-03-15 is +\$14.61 billion on the published inputs and +\$14.61 billion as call-only GEX, −\$4.51 billion without same-day expiries and −\$15.35 billion on the complete series. The archive value is +\$12.62 billion and is put-less too, so the step from the archive to the rebuilt series cannot be read as an effect of the gamma model alone: it combines the restored put leg with the change of the gamma model.

Table 8. 0DTE ablation, SPY, by year: share of the absolute net GEX contributed by same-day expiries (percent), and detection rates (percent) of the classifier and of gpt-6-luna on `new_full` and `new_ex0dte`. The McNemar p-values are exact tests on the non-overlapping subsample (Section 3.8), given where the result files hold the test for the period (2024, 2025, 2026 and the pooled periods); the block-bootstrap values accompany them in `results/stats.yaml`.

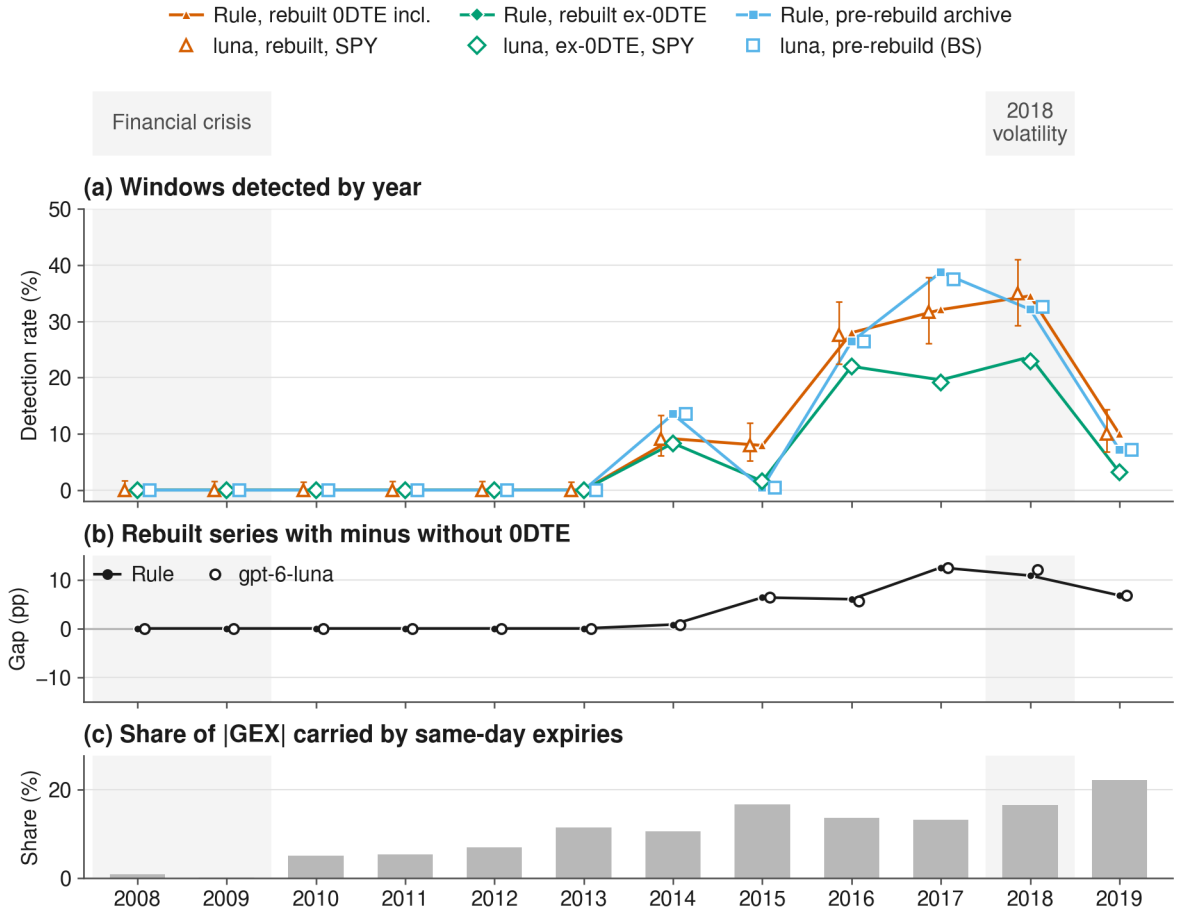
	0DTE share	Classifier		gpt-6-luna		McNemar p	
Year	of |GEX|	full	ex-0DTE	full	ex-0DTE	classifier	gpt-6-luna
2020	25.7	22.2	10.7	22.2	10.7	–	–
2021	36.0	3.6	0.0	3.6	0.0	–	–
2022	23.4	40.6	36.7	41.8	37.1	–	–
2023	35.8	22.4	25.2	23.2	26.0	–	–
2024	45.9	10.3	13.1	11.1	14.3	1.000	1.000
2025	43.9	13.6	16.4	14.0	16.4	1.000	1.000
2026	40.8	11.4	21.2	11.4	21.2	1.000	1.000
2008–2019 (pooled)	–	10.1	6.5	10.1	6.4	0.063	0.063
2020–2023 (pooled)	–	22.2	18.1	22.7	18.4	1.000	0.688
2024–2025 (pooled)	–	12.0	14.7	12.5	15.3	1.000	1.000

4.6. Deep Baseline, 2008–2019

Figure 8 extends the analysis to 2008–2019, before the growth of weekly and daily expirations. Before 2014 no window of any of the three series is detected, by the rule or by gpt-6-luna; this includes the 474 windows that end in the financial crisis of 2008 and 2009. Detections start in 2014 (9.1% on `new_full`) and peak in 2018 on both complete series, at 34.5% on `new_full` (86 of 249 windows) and 23.7% on `new_ex0dte`, the year of the 2018 volatility episode; the archive peaks in 2017 (38.8%). Pooled over 2008 to 2019 the rule detects 10.1% of the windows of `new_full` (301 of 2,971; Wilson 9.1–11.3%, quarter-block bootstrap 4.9–16.6%), 6.5% of those of `new_ex0dte` and 9.8% of those of the archive. The model gives 10.1%, 6.4% and 9.8% on the same three series.

The gap between `new_full` and `new_ex0dte` is a consistency check of the ablation arm. It is zero from 2008 to 2013, although same-day expiries carry 11.5% of the absolute net GEX in 2013, and it is positive in every year from 2014 to 2019: 0.8 percentage points in 2014, 6.3 in 2015, up to 12.5 in

2017 and 6.8 in 2019, while the share rises to 22.1%. The gap is therefore not proportional to the share. The model reproduces it (up to 12.5 percentage points). Over the whole period the complete series detects 3.6 percentage points more than the ex-0DTE series (119 windows detected only with same-day expiries against 12 only without; quarter-block bootstrap $p = 5.0 \times 10^{-4}$; McNemar on the non-overlapping subsample of 100 windows $p = 0.063$). Same-day expiries thus raise the detection rate in the decade before 0DTE, whereas the sign of the difference is reversed, without a significant difference, in 2024 and 2025 (Section 4.5).



SPY, 30-day windows ending in the year. Lines: deterministic rule; open markers: gpt-6-luna on the same windows (shifted sideways for legibility). Whiskers on the first series: Wilson 95% interval, which treats the overlapping windows of a year as independent and is too narrow: a year holds about 9 non-overlapping 30-day windows. (b) Detection rate with 0DTE minus without, on the windows both series have. Shaded: 2008–2009 financial crisis and 2018 volatility episode.

Figure 8. Pre-0DTE baseline, SPY, 2008–2019, with the 2008–2009 financial crisis and the 2018 volatility episode marked. (a) Detection rate by year of the classifier (lines) and gpt-6-luna (markers). (b) Gap between `new_full` and `new_ex0dte`. (c) Share of |GEX| carried by same-day expiries. **Read this figure as:** all three series stay at zero until 2013 and diverge as the same-day share in panel (c) rises, with the largest gap of 12.5 percentage points in 2017. The markers of gpt-6-luna lie close to the lines of the rule in every year.

4.7. Cross-Asset Replication and the Floor Effect

Table 9 and Figure 9 repeat the analysis for QQQ and IWM. The fixed dollar thresholds of Section 3.1 can produce a floor effect for smaller underlyings, which we report; the scaled series of Section 3.6 show whether the model's verdict still follows the criteria when the floor is lifted. The floor effect is complete. Under the fixed thresholds the rule detects no window of QQQ (0 of 3,867, windows ending 2011 to 2026-10-09) and none of IWM (0 of 4,678, 2008 to 2026-10-09), in any year and on either series. The largest 30-day mean |GEX| is $4.92 billion for QQQ and $4.98 billion for

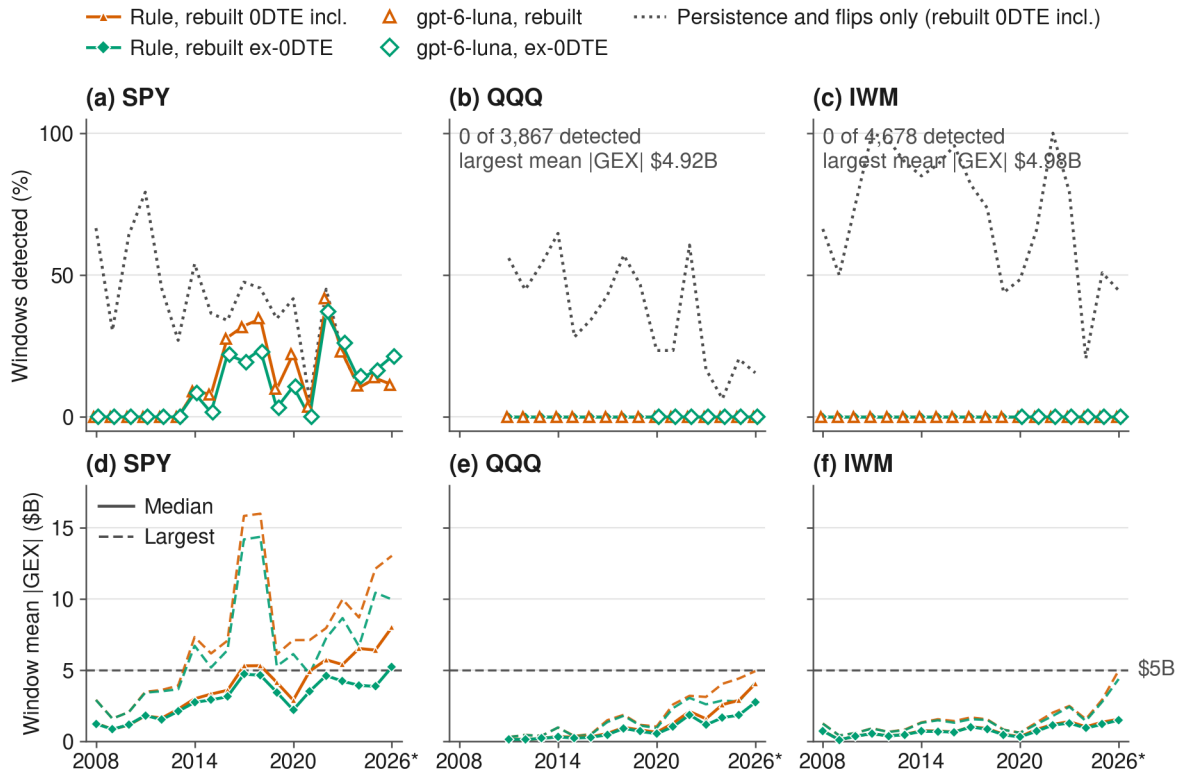
IWM on `new_full` ($4.11 billion and $4.37 billion without same-day expiries), against $16.00 billion for SPY, so no window reaches the $5 billion criterion. This is a consequence of the fixed thresholds, a pre-registered design decision, and not a modelling result. The magnitude criterion is the binding one: the persistence and sign-flip tests alone pass 1,434 of 3,867 windows of QQQ (37.1%) and 3,357 of 4,678 of IWM (71.8%), against 1,763 of 4,671 windows of SPY (37.7%). gpt-6-luna labels no window of QQQ or IWM as persistent either (0 of 3,867 and 0 of 4,678 on `new_full`; 0 of 1,699 and 0 of 1,700 on `new_ex0dte`), so the unscaled arms show that the model also rejects every window; they cannot test its discrimination.

The scaled variant multiplies the `new_full` values by one factor per symbol: 4.225 for QQQ (median of 1,699 daily ratios) and 11.431 for IWM (1,700 days). It is a disclosed protocol change (Section 3.6). With it the rule detects 350 of 1,670 windows of QQQ (21.0%) and 901 of 1,671 windows of IWM (53.9%), and gpt-6-luna 360 (21.6%) and 904 (54.1%). The model follows the rule on the scaled inputs: the verdicts agree in 99.4% of the QQQ windows (kappa 0.982) and 99.7% of the IWM windows (kappa 0.994). The paired difference to the unscaled arms is 21.0 percentage points for QQQ (quarter-block bootstrap interval [12.6, 31.2], $p = 5.0 \times 10^{-4}$; exact McNemar test on the non-overlapping subsample $p = 4.9 \times 10^{-4}$) and 53.9 for IWM ([40.7, 67.4], $p = 5.0 \times 10^{-4}$; $p = 4.7 \times 10^{-10}$); for the model the differences are 21.6 and 54.1 percentage points.

The scaled rates depend on the factor and are not a measurement of QQQ or IWM regimes. The ratio of the open-interest notional of SPY to that of the symbol is not constant: for QQQ the daily ratio runs from 0.78 to 38.32 (quartiles 3.30 and 5.73), and for IWM from 1.51 to 110.57 (quartiles 7.72 and 16.46). A single factor puts the exposure on the scale of SPY on a typical day, no more, and another factor would give other rates. The scaled arms therefore show only that the model follows the criteria once the floor is lifted. They show no step after 2023: the rate of the rule on the scaled IWM series is 65.2% in 2020 to 2023 and 35.7% in 2024 and 2025 (shift -29.5 percentage points, $p = 0.061$ by bootstrap, $p = 0.137$ by Fisher exact test on the subsample), and on the scaled QQQ series 25.9% and 13.4% (-12.5 percentage points, $p = 0.170$, $p = 0.292$).

Table 9. Cross-asset replication: windows, typical magnitude and persistent windows under the fixed thresholds, by symbol and series. The scaled series are the disclosed protocol change of Section 3.6. The magnitude column is the median over windows of the 30-day mean |GEX| of the series as sent to the model; the scaled series are SPY-scale values, so their magnitudes are not comparable with the unscaled rows of the same symbol. The ex-0DTE arms of QQQ and IWM start in 2020, so gpt-6-luna answered fewer windows than the classifier labelled.

Symbol	Series	Windows	Median mean |GEX| ($B)	Persistent (classifier)	Detected (gpt-6-luna)
SPY	`new_full`	4,671	3.84	606 of 4,671 (13.0%)	613 of 4,671 (13.1%)
QQQ	`new_full`	3,867	0.74	0 of 3,867 (0.0%)	0 of 3,867 (0.0%)
IWM	`new_full`	4,678	0.66	0 of 4,678 (0.0%)	0 of 4,678 (0.0%)
SPY	`new_ex0dte`	4,671	3.04	491 of 4,671 (10.5%)	494 of 4,671 (10.6%)
QQQ	`new_ex0dte`	3,867	0.64	0 of 3,867 (0.0%)	0 of 1,699 (0.0%)
IWM	`new_ex0dte`	4,678	0.62	0 of 4,678 (0.0%)	0 of 1,700 (0.0%)
QQQ	`new_full_scaled`	1,670	9.13	350 of 1,670 (21.0%)	360 of 1,670 (21.6%)
IWM	`new_full_scaled`	1,671	12.00	901 of 1,671 (53.9%)	904 of 1,671 (54.1%)



30-day windows ending in the year (QQQ from 2011; ex-0DTE series of QQQ and IWM from 2020). Row 1: lines are the deterministic rule, open markers gpt-6-luna on the same windows; the dotted line is the share of windows that meet the persistence (21 of 30 days) and sign-flip (5) tests, whatever their magnitude; the notes count the rebuilt 0DTE-incl. series. Row 2: median and largest mean |GEX| of the windows ending in the year; the dashed horizontal line is the fixed $5B magnitude threshold. * 2026: windows ending up to 2026-10-09.

Figure 9. Cross-asset replication, SPY, QQQ and IWM (QQQ from 2011). Top row: detection rate by year of the classifier (lines) and gpt-6-luna (markers) on `new_full` and `new_ex0dte`, and the share of windows that pass the persistence and sign-flip tests alone (dotted). Bottom row: yearly median and largest 30-day mean |GEX| against the $5 billion criterion. **Read this figure as:** for QQQ and IWM the detection lines lie at zero in every year while the dotted lines show that many windows pass the persistence and flip tests; the bottom row shows why, since the largest yearly window mean stays below the $5 billion line for QQQ (at most $4.92 billion) and for IWM (at most $4.98 billion). For SPY the largest window mean reaches $16.00 billion.

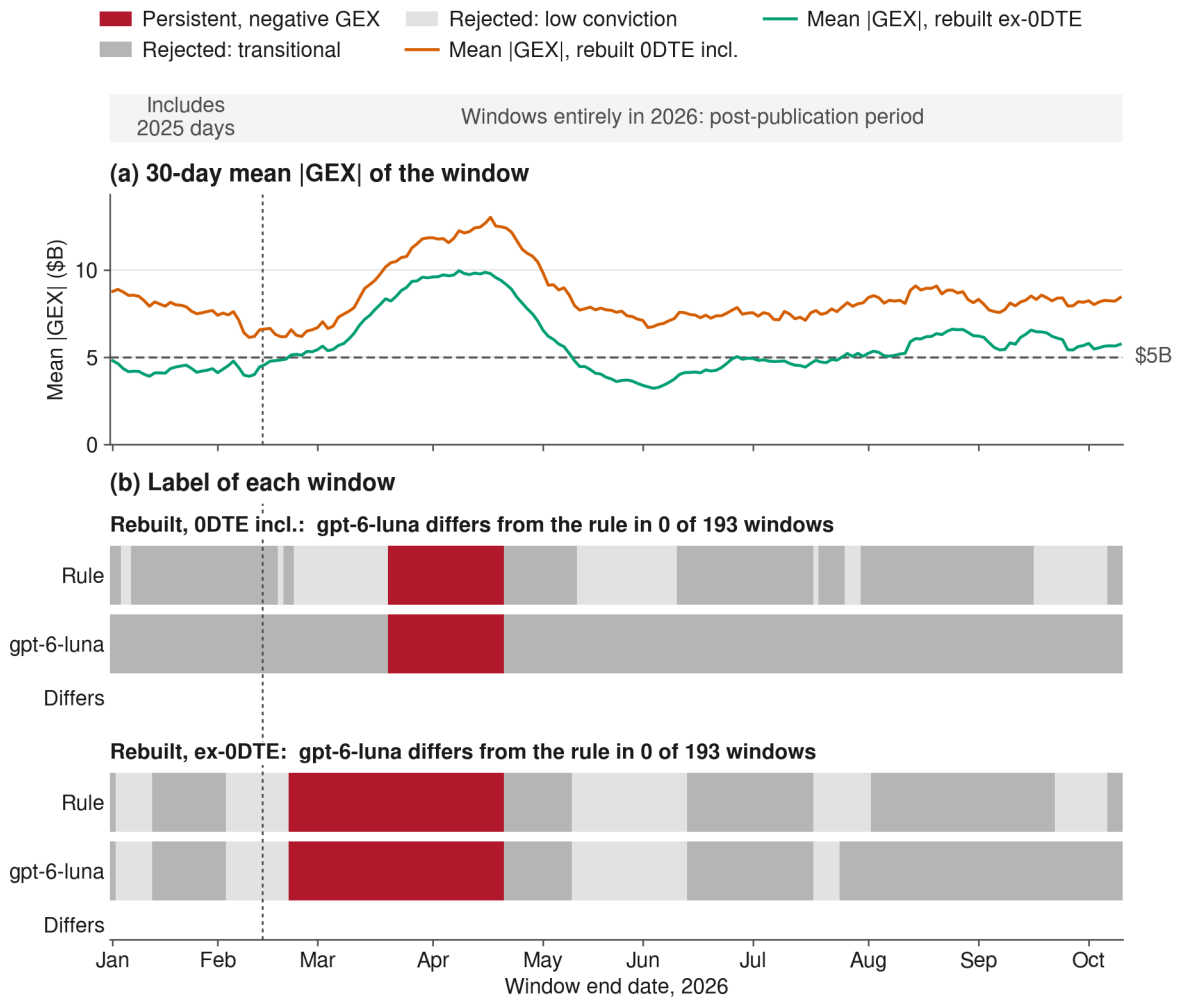
4.8. Post-Publication Period

Table 10 and Figure 10 cover the post-publication period of Section 3.4: the windows that end in 2026, of which those with ends from 2026-02-13 lie entirely after the end of the published sample (2025-12-31). We use this label rather than out-of-sample, because gpt-6-luna was created on 2026-09-14 and its training cutoff is undocumented. The period has 193 windows with ends from 2026-01-02 to 2026-10-09, of which 164 lie entirely in 2026 (ends from 2026-02-13) and 29 still contain days of 2025. The last window end is 2026-10-09 because equity options moved from Alpha Vantage to Cboe delayed quotes on 2026-10-12 (Section 3.3). On the complete series the rule detects 22 of 193 (11.4%) of the windows that end in 2026 and 22 of 164 (13.4%) of those entirely in 2026; without same-day expiries it detects 41 of 193 (21.2%) and 41 of 164 (25.0%). All detected windows of the complete series are persistent negative-GEX windows, with ends between 2026-03-20 and 2026-04-21. These rates are close to the 12.0% of 2024 and 2025 on the complete series and far from the 100% that the published inputs gave for those years. The Wilson interval for the windows ending in 2026 is 7.6–16.7% (quarter-block bootstrap 0.0–22.1%, 4 blocks). A call-only series would label 193 of the 193 windows as persistent, with a mean |GEX| of $21.73 billion. gpt-6-luna agrees with the rule in 100.0% of the windows on `new_full` and 100.0% on `new_ex0dte` (0 and 0 windows of disagreement, respectively). For QQQ and

IWM the unscaled series give no detection by the rule or by the model (0 of 193 windows each). On the scaled series the rule detects 15.5% (QQQ) and 44.6% (IWM) of the windows ending in 2026 and the model 16.1% and 45.1%. The training data of gpt-6-luna may overlap this period, so the agreement of the model with the rule here shows consistency with the rule and says nothing about forecasting.

Table 10. Post-publication period, SPY, windows ending in 2026 up to 2026-10-09 (those with ends from 2026-02-13 lie entirely in 2026). Cells give windows detected, with the percentage in brackets.

Series	Window set	Windows	Classifier detects (%)	gpt-6-luna detects (%)	Agreement (%)
`new_full`	ends in 2026	193	22 (11.4)	22 (11.4)	100.0
`new_full`	entirely in 2026	164	22 (13.4)	22 (13.4)	100.0
`new_ex0dte`	ends in 2026	193	41 (21.2)	41 (21.2)	100.0
`new_ex0dte`	entirely in 2026	164	41 (25.0)	41 (25.0)	100.0



Each cell is a 30-day window ending on that trading day (193 window ends). The dotted line marks the first window that lies entirely in 2026; earlier windows still contain days of 2025. Dashed horizontal line in (a): the fixed $5B magnitude threshold.

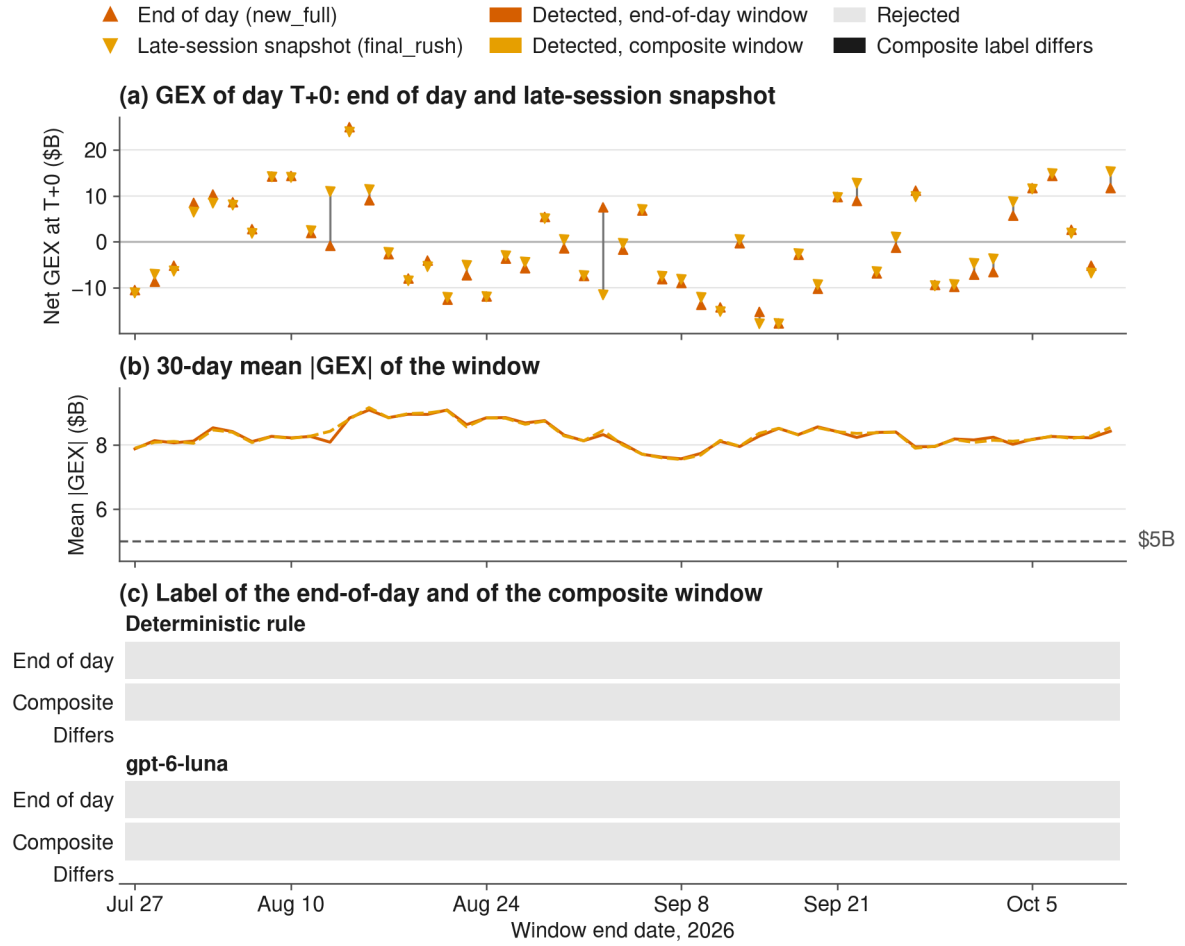
Figure 10. Post-publication period (windows ending in 2026 up to 2026-10-09): per-window classifier and gpt-6-luna labels for SPY under the complete series; the windows that still contain days of 2025 are set apart. **Read this figure as:** the 30-day mean |GEX| of the complete series (a) stays above $5 billion in every window (lowest value $6.13 billion), whereas that of the ex-0DTE series is below it in 84 of 193 windows. The rule labels 22 windows of the complete series and 41 windows of the ex-0DTE series as persistent (red in b), and gpt-6-luna differs from the rule in 0 of the 193 windows of the complete series and in 0 of the 193 windows of the ex-0DTE series.

4.9. Intraday Check

Table 11 and Figure 11 compare the late-session intraday snapshot with the end-of-day value for SPY from 2026-07-27 to 2026-10-09. The snapshot is the `final_rush` slot (19:55:59 UTC), the last independent late-session observation; the `market_close` slot equals the end-of-day row. The irregular days 2026-07-29, 2026-08-03, 2026-09-14 are excluded, which leaves 51 days. The two values of day T+0 have the same sign on 46 of 51 days (90.2%). Their median relative difference is 11.9% and the median absolute difference is \$0.93 billion (mean \$1.69 billion, largest \$19.08 billion). A single day moves the 30-day mean |GEX| of a window only a little: the composite window (the complete series for days $T-29$ to $T-1$ and the snapshot at T+0) differs from the end-of-day window by at most \$0.34 billion. The rule gives the same label for the composite and the end-of-day window on 51 of 51 days, and gpt-6-luna on 51 of 51. This agreement is weak evidence, because no window of the period is persistent: the rule and the model detect 0 and 0 of the 102 windows, and the 30-day mean |GEX| of every window is at least \$7.54 billion, so the verdict depends on persistence and flips, not on the magnitude. With a few dozen days the section reports counts and no p-values.

Table 11. Late-session intraday snapshot versus end-of-day GEX, SPY, 2026-07-27 to 2026-10-09.

Statistic	Value
Days with both observations	51
Days with the same sign (%)	90.2 (46 of 51)
Median relative difference of GEX (%)	11.9
Median absolute difference of GEX (\$B)	0.93
Classifier label agreement of composite and end-of-day windows (%)	100.0 (51 of 51)
gpt-6-luna label agreement of composite and end-of-day windows (%)	100.0 (51 of 51)



SPY, 51 composite window ends. On the 51 regular days the two T+0 values have the same sign on 46; the median relative difference is 11.9%. The rule gives the same label for the composite and the end-of-day window on 51 of 51 days, gpt-6-luna on 51 of 51. Dashed line in (b): the fixed $5B threshold.

Figure 11. Late-session intraday snapshot versus end-of-day GEX, SPY, 2026-07-27 to 2026-10-09. (a) Day T+0 value of the snapshot (`final_rush`) and of the end-of-day row. (b) 30-day mean |GEX| of the composite and of the end-of-day window against the $5 billion criterion. (c) Labels of the classifier and of gpt-6-luna on the two windows. The three irregular days (2026-07-29, 2026-08-03, 2026-09-14) are not in the series. **Read this figure as:** the snapshot and the end-of-day value of day T+0 are close on most days and differ in sign on 5 of 51 days. The two window means in (b) are almost the same line, and in (c) every window is rejected under both labels, with no day on which the composite label differs.

4.10. Model Fidelity to the Rule

Table 12 and Figure 12 compare each model answer with the deterministic classifier on the same window. The verdict of the model equals the verdict of the rule in 99.6% or more of the windows of every SPY arm (99.7% on `full_spy`, 99.7% on `ex0dte_spy`), with kappa between 0.985 and 0.995. The published o4-mini answers equal the rule in all 1,412 windows, and gpt-6-luna reaches 99.4% and 99.7% on the scaled QQQ and IWM series. On the unscaled QQQ and IWM arms the accuracy is 100.0% and kappa is undefined, because the rule labels no window as persistent. Over the answers drawn in panels (c) to (e) of Figure 12 (33,160 parsed answers) the verdict equals the rule's in 99.8% of the 31,684 gpt-6-luna answers and in 100.0% of the 1,476 o4-mini answers.

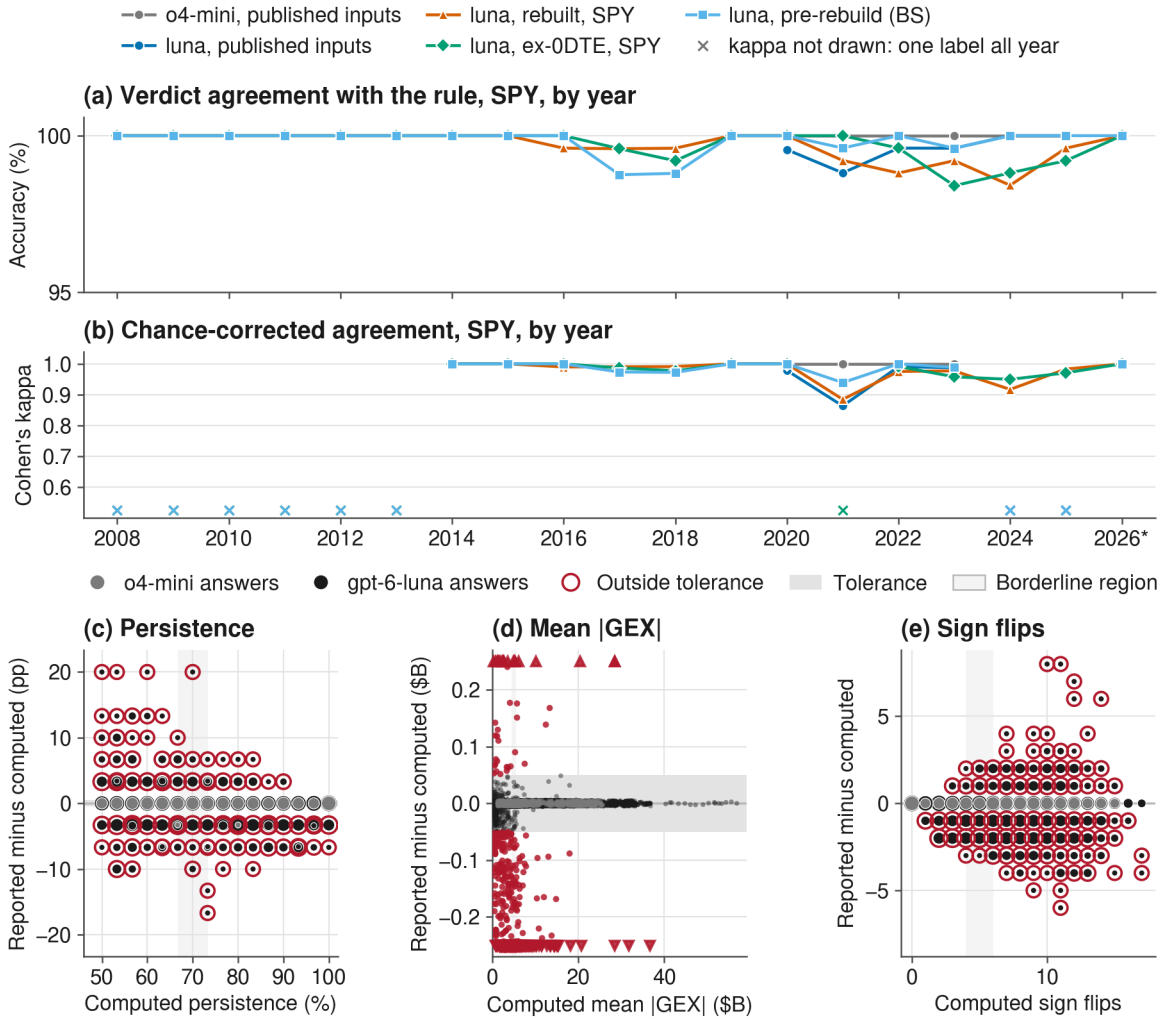
The reported metrics are less exact than the verdict, and least exact on the unscaled QQQ and IWM arms (82.2% to 89.5%). The share of answers whose reported counts, magnitude and persistence equal the computed values (within 0.5 days, $0.05 billion and 0.5 percentage points) is 99.4% for o4-mini and between 92.7% and 98.1% for gpt-6-luna on the SPY arms. The sign-flip count is exact in 100.0% of the o4-mini answers and in 93.2% to 95.6% of the gpt-6-luna answers on the SPY arms

(93.3% over all gpt-6-luna answers of Figure 12); the model undercounts flips in 6.0% of its answers and overcounts them in 0.6%. Its reported persistence lies within the tolerance of the figure (one day) in 94.9% of the answers and its reported magnitude within 10% of $5 billion in 98.7%, against 99.9% and 99.6% for o4-mini. The median ratio of reported to computed magnitude is 1.000.

The few verdicts that differ from the rule lie at the thresholds. Seven gpt-6-luna arms have windows in which the verdict differs from the rule (`full_spy`, `ex0dte_spy`, `full_qqq_scaled`, `bridge_old`, `bridge_jan`, `full_iwm_scaled`, `p34_obf_2024`), 58 windows in all (`results/disagreements.csv`): 39 are windows in which the model reports at most five sign flips where the rule counts six or more, so that it calls a window persistent that the rule classifies as low conviction; 15 are windows in which the reported mean magnitude lies on the other side of $5 billion than the computed one; 2 are overcounted flips; and 2 are windows whose reported values are correct and meet all three criteria but which the model labels transitional (persistence 70.0% and 73.3%, close to the 70% threshold). In 41 of the 58 windows the model detects a regime that the rule rejects, and in 17 it rejects one that the rule detects. In 51 of the 58 windows the computed values lie in the borderline region of Figure 12, that is a persistence within one day of 21 of 30 days (66.7–73.3%), a mean magnitude between $4.50 and $5.50 billion or 4 to 6 flips. Away from these regions the model reproduces the rule.

Table 12. Fidelity of model answers to the deterministic classifier on the same windows, by arm: accuracy of the binary verdict, Cohen's kappa (– where the classifier or the model labels every window alike), the share of answers whose reported counts, magnitude and persistence equal the computed values (0.5 day, $0.05 billion, 0.5 percentage points), and the share with an exact sign-flip count. The published o4-mini answers of `r1_o4mini`, `r1_p3_o4mini` and `r1_p4_o4mini` form the first block.

Arm	Windows	Accuracy (%)	Cohen's kappa	Metrics agree (%)	Flips exact (%)
`r1_o4mini`	1,412	100.0	1.000	99.4	100.0
`r1_p3_o4mini`	223	100.0	1.000	100.0	100.0
`r1_p4_o4mini`	223	100.0	1.000	100.0	100.0
`bridge_jan`	1,476	99.6	0.992	98.1	95.2
`bridge_old`	4,638	99.8	0.995	94.2	95.6
`full_spy`	4,671	99.7	0.986	92.7	93.2
`ex0dte_spy`	4,671	99.7	0.985	93.4	94.6
`full_qqq`	3,867	100.0	–	82.2	89.6
`full_iwm`	4,678	100.0	–	85.9	92.6
`ex0dte_qqq`	1,699	100.0	–	89.5	92.2
`ex0dte_iwm`	1,700	100.0	–	86.0	89.2
`full_qqq_scaled`	1,670	99.4	0.982	93.5	93.2
`full_iwm_scaled`	1,671	99.7	0.994	97.5	95.8
`p34_obf_2024`	223	99.6	0.985	98.2	96.4
`p34_dated_2024`	223	100.0	1.000	99.6	98.2
`p34_obf_2020`	223	100.0	1.000	100.0	100.0
`p34_dated_2020`	223	100.0	1.000	100.0	100.0
`intraday_spy`	51	100.0	–	96.1	92.2



(c)–(e): 33,160 parsed answers of all registered arms; reported minus the value the rule computes for the same window, against the computed value (marker area grows with the log of the number of identical answers in (c) and (e)). Light band: within one day of the 21-day persistence threshold, within 10% of $5B, within one flip of the limit of 5. Reports inside the tolerance (persistence, mean |GEX|, exact sign flips): o4-mini 99.9%, 99.6%, 100.0%; luna 94.9%, 98.7%, 93.3%. In (d), 205 errors beyond ±$0.25B sit on the frame as triangles. Accuracy on QQQ and IWM: full_qqq 100.0%, ex0dte_qqq 100.0%, full_iwm 100.0%, ex0dte_iwm 100.0%. * 2026: windows ending up to 2026-10-09.

Figure 12. Fidelity of the model answers (o4-mini and gpt-6-luna) to the deterministic classifier. (a), (b) Accuracy and Cohen's kappa of the verdict by series and year. (c)–(e) Reported versus computed persistence, mean |GEX| and sign-flip count, with the borderline region around each threshold shaded. **Read this figure as:** the yearly accuracy stays between 98.4% and 100% for every arm and year, (lowest value: `ex0dte_spy` in 2023). In panels (c) to (e) the o4-mini answers (grey) lie on the zero line, whereas the gpt-6-luna answers (black) scatter around it; the errors that matter for the verdict are those inside the shaded regions, and the largest scatter is in the sign-flip count, where the model undercounts.

4.11. Dated versus Obfuscated Prompts: The 2024 and 2020 Baselines

Table 13 compares the published o4-mini answers on the inputs of the Phase 3 and 4 baselines, whose prompts carried calendar dates, with gpt-6-luna on the same values with the dates and with relative day labels (Section 3.6). The published o4-mini answers detect 181 of 223 (81.2%) of the 2024 windows and 27 of 223 (12.1%) of the 2020 windows (Wilson 95% intervals 75.5–85.8% and 8.5–17.0%). With the dates, gpt-6-luna reproduces these answers: it gives the same verdict in 223 of 223 windows of 2024 and 223 of 223 of 2020 (kappa 1.000 and 1.000). The deterministic classifier on the series as sent gives the same labels as o4-mini in both baselines (100.0% and 100.0% agreement).

Showing the dates changed nothing that the data can detect. With relative day labels gpt-6-luna detects 180 of 223 (80.7%) of the 2024 windows against 181 of 223 (81.2%) with dates, and 27 of

223 (12.1%) of the 2020 windows against 27 of 223 (12.1%). The two labelings differ in 1 window of 2024 and in none of 2020 (paired quarter-block bootstrap interval for 2024 $[-1.9, 0.0]$ percentage points, $p = 0.646$; exact McNemar test $p = 1.000$ at stride one and $p = 1.000$ on the non-overlapping subsample of 8 windows; in 2020 both are $p = 1.000$). The 223 windows of each baseline overlap with stride one, and the quarter-block bootstrap rests on 4 calendar quarters, so these intervals are coarse.

The 2024-versus-2020 contrast of the earlier studies is reproduced under every label type. The difference in detection rate between the two baselines is 69.1 percentage points for the published o4-mini answers ($\varphi = 0.692$, Newcombe–Wilson interval [61.6, 74.9]; quarter-block bootstrap interval [31.6, 93.3], $p = 0.002$; Fisher exact test on the non-overlapping subsample of 8 windows per baseline $p = 0.041$), 69.1 for gpt-6-luna with dates ($\varphi = 0.692$) and 68.6 for gpt-6-luna with relative labels ($\varphi = 0.688$; bootstrap $p = 0.003$; Fisher $p = 0.041$). The Fisher test on 8 non-overlapping windows per baseline is far less extreme than the test on all windows, which treats the overlapping windows as independent ($p = 1.8 \times 10^{-52}$).

The contrast is one of magnitudes and not of regimes. The two input series are not on one convention: the 2024 values are a gross magnitude with an imposed negative sign and a mean absolute value of \$31.24 billion per day, the 2020 values are a magnitude without sign and have a mean of \$2.85 billion (Section 4.1). The 30-day mean $|\text{GEX}|$ of the windows averages \$30.71 billion in 2024 and \$2.56 billion in 2020, against the criterion of \$5 billion, so the contrast does not describe a negative against a positive regime, and the type of day label does not change it.

Table 13. Phase 3 (2024) and Phase 4 (2020) baselines: detection rate in percent (detected windows out of 223 in brackets) of the published o4-mini answers, which carried dates, of gpt-6-luna with the dates and with relative day labels (Day T), and of the deterministic classifier on the series as sent. The last column is the exact McNemar p-value of the two gpt-6-luna arms on the non-overlapping subsample (Section 3.8). The last row gives the difference between the two baselines in percentage points with φ in brackets (unpaired).

Baseline	Series	o4-mini, dated	luna, dated	luna, Day T	Classifier	McNemar p
Phase 3, 2024	`p3_2024`	81.2 (181)	81.2 (181)	80.7 (180)	81.2 (181)	1.000
Phase 4, 2020	`p4_2020`	12.1 (27)	12.1 (27)	12.1 (27)	12.1 (27)	1.000
2024 minus 2020, pp (φ)		69.1 (0.692)	69.1 (0.692)	68.6 (0.688)	69.1 (0.692)	–

5. Discussion

5.1. What Survives

The data audit leaves intact the finding that detection is rare before 2024 and varies from year to year. In 2021 to 2023 the rule gives 4.0%, 32.3% and 19.3% on the published inputs and 3.6%, 40.6% and 22.4% on the complete series, and the model reproduces the rule on both. The audit removes the saturation of 2024 and 2025. The published inputs give 100.0% in both years because they contain the call leg only; the complete series gives 10.3% and 13.6% (Section 4.2). The period contrast of 2020 to 2023 against 2024 and 2025 is a rise of 83.0 percentage points on the published inputs and a change of -10.2 points on the complete series, a change that the data do not distinguish from zero (bootstrap $p = 0.108$). The lineage finding bears on the 0DTE reading of the 2023-to-2024 shift. Same-day expiries carry 45.9% of the absolute net GEX of SPY in 2024, against 22.1% in 2019, so the growth of 0DTE trading is visible in the exposure itself. The sudden step of the detection rate between 2023 and 2024 is not: it coincides with the change of the input from the complete series to the call leg (Section 4.1), and it does not appear on the complete series. Removing same-day expiries does not restore it (12.0% with and 14.7% without them in 2024 and 2025; Section 4.5). We therefore read the published rise as a property of the published inputs. The complete series supports only the weaker statement that same-day expiries form a large part of the exposure, and it claims no mechanism that links them to the

regime labels. The years of the replay are not independent observations, so these readings rest on the overlap-aware checks of Section 3.8 and not on the Wilson intervals.

5.2. What the Model Tracks

The verdict of the model follows the numbers it is shown and the criteria in the prompt. In the probe of 2026-10-28 it equalled the rule in 31 of 31 responses, and in the full runs it equals the rule in 99.7% of the windows of both complete SPY arms (kappa 0.986 and 0.985). The few verdicts that differ lie at the thresholds: 58 windows over seven arms, 51 of them in the borderline region of persistence, magnitude or flip count (Section 4.10). Obfuscation therefore validates the model on a given series. It does not show that the series is what the prompt says it is. The same models, prompts and rule give 100.0% or about 12% on the same years, depending only on which series enters the prompt, so the provenance of the input series is part of the test. The rerun of the Phase 3 and 4 baselines points the same way for the visibility of dates. With calendar dates gpt-6-luna reproduces the November 2025 answers in 223 of 223 windows in each baseline, and with relative day labels it differs in 1 window of 2024 and in none of 2020 (Section 4.11). Visible dates changed nothing that these data can detect. The model is sensitive to the values and the criteria, and it shows no sensitivity to the calendar that the replay can measure.

5.3. Threats to Validity

All model arms use one provider and one model family, so the agreement with the rule may differ for other models; the alias gpt-6-luna has no dated snapshot, and we store the creation time of the model object and the response identifier of every answer instead. Equity options moved from Alpha Vantage to Cboe delayed quotes on 2026-10-12, so the last window ends on 2026-10-09, and the end-of-day open interest differs from the intraday value (the sign of the net GEX is the same on 46 of 51 days, 90.2%, and the check rests on a single snapshot type, the `final_rush` slot). The prompt shows values rounded to 0.01B whereas the rule uses the unrounded values, which can move a window across a threshold. Windows have stride one and share 29 of their 30 days, so the Wilson intervals are optimistic, and we rely on the quarter-block bootstrap, which has few blocks per year, and on the non-overlapping subsample, which has few windows (Section 3.8). The thresholds come from the earlier work and are fixed in dollars for every symbol; for QQQ and IWM they produce a floor effect (no window is detected in 3,867 and 4,678 windows) that the unscaled arms cannot test. Each window is answered once unless an arm was repeated; the repeat of the bridge arm differs from the first pass in 11 windows (99.3% agreement), so a single run carries a small amount of sampling noise at the thresholds. The model undercounts sign flips in 6.0% of its answers, which is the main source of the verdicts that differ from the rule, and the stored count of the rule is the reference. The post-publication period is short (193 windows ending in 2026, of which 164 lie entirely after 2025) and is not out-of-sample for the model, whose training data may cover it; the agreement there shows consistency with the rule and no predictive skill. The raw option chains come from licensed vendors and are not redistributed; the condensed per-window tables, which hold the inputs, the labels and the answers, are the basis for reproduction (Section 3.7). The Phase 3 and 4 baselines are replayed verbatim, including two sign conventions and ten placeholder rows, and the scaled QQQ and IWM variants use one factor per symbol, a disclosed protocol change that was decided after the thresholds were fixed; their rates depend on the factor and do not measure QQQ or IWM regimes.

5.4. Implications

For the published results, the relevant facts are these. The published 2024 and 2025 inputs were the call leg of the options chain from 2023-09-29, the labels follow from those inputs and from the published criteria, and the complete series gives other rates for the same years. A data section that states the input convention and the date of the rebuild of the daily table would let a reader place a

result on the right series. The decision whether and how to add such a statement to the earlier papers rests with the authors.

For the validation of AI-enabled models in general, the study suggests four practices. Inputs should be versioned and stored with each model run, together with the labels and the raw answers, so that every verdict can be traced to its values. A lineage check between runs should compare the inputs of a new run with those of the previous run before the labels are compared. Drift alarms on upstream rebuilds should flag a change in the construction of an input table. A bridge run on identical inputs should accompany the retirement of a model, as the bridge between o4-mini and gpt-6-luna does here (kappa 0.993 on 1,412 windows). These practices address the theme of the Special Issue, the assurance of AI-enabled electronic systems, at the point where an upstream data change reaches the model without any change to the model.

6. Conclusions

This study re-runs the detection of persistent dealer-gamma regimes on the complete data. The published 2024 and 2025 inputs were the call leg of the options chain from 2023-09-29; on them the rule detects 100.0% of the windows of both years, and on the complete series it detects 12.0% (60 of 502; quarter-block bootstrap 5.0–19.2%). The regime labels are therefore sensitive to the input convention, and the earlier saturation is a property of the inputs; the complete series shows no increase after 2023 that the data distinguish from zero. A second model, gpt-6-luna, agrees with the published o4-mini answers on 99.6% of 1,412 windows (kappa 0.993), follows the rule on the complete series in 99.7% of the windows, and is unchanged by dates in the prompt (1 window of 223 differs in the 2024 baseline). The scope is limited to one provider, one model alias, fixed dollar thresholds and overlapping windows, and the scaled QQQ and IWM variants and the single intraday snapshot are checks and not measurements. Future work includes a scaled-threshold variant, which changes the prompt text and is a new prompt version, intraday open interest, other models and vendors, and standards for versioned inputs in LLM validation studies.

Appendix A. Prompt, Schema and API Configuration

This appendix reproduces the complete request sent to the model for each 30-day window, so that the experiment can be repeated from the manuscript alone. The prompt text is the output of `MechanicsPromptBuilder.build_regime_prompt` (file `src/llm/mechanics_prompt_builder.py`); the schema and the request body are the constant `REGIME_JSON_SCHEMA` and the function `chat_body` of `scripts/validation/electronics/common.py`. The listings were generated from these objects and compared with them.

Appendix A.1. Model and API Configuration

- **Model:** `gpt-6-luna`, an OpenAI model alias without a dated snapshot. The creation timestamp of the model object (2026-09-14) is stored with every run and the response identifier with every answer. The published run used `o4-mini-2025-04-16`.
- **Access:** OpenAI Batch API, endpoint `/v1/chat/completions`, completion window of 24 hours, at most 1,500 requests per batch job. Each input line is one request that carries the key `<arm>|<symbol>|<window_end>` in `custom_id`; a test-retest duplicate adds the suffix `#rep1`, and its arm is stored as `<arm>#rep1`.
- **Reasoning effort:** `medium`, the default effort of o4-mini in the published run.
- **Maximum completion tokens:** 4,000; reasoning tokens count against the limit. An answer that ends with `finish_reason` = `length` is requested once more with 8,000.
- **Response format:** the strict JSON schema of Appendix A.3. The earlier runs requested the same nine fields in the prompt text only; this is the one protocol change of the study.
- **Not sent:** temperature, seed, system message, tools and stop sequences.

- **Parsing:** an answer is parsed as JSON; a second attempt tolerates raw control characters inside strings, and a last fallback extracts the JSON object from free text. The parse status of every answer (`ok`, `fallback`, `failed` or `missing`) is stored.
- **Prices:** USD 0.05 per million input tokens and USD 0.25 per million output tokens on Batch, twice that on the standard API.

Identical requests need not return identical answers, because no seed is set. The test-retest duplicates of Section 3.6 measure this variation, and the deterministic classifier, not the model, supplies the reference label of each window.

Appendix A.2. Prompt Text

The prompt is a single user message. The placeholder `{gex_data_table}` stands for the 30 data lines of a window, one per day, in the form `Day T-29:  +3.42B` to `Day T+0:  -12.18B`: a relative day label, the net GEX in billions of US dollars with two decimals and an explicit sign, and neither a date nor a ticker. In the two dated arms (`p34_dated_2024` and `p34_dated_2020`) the label is instead the ISO date of the trading day, as in the November 2025 runs (`2024-01-02:  -33.01B`). The rest of the text is the same for every window, series and symbol (prompt version `v2.0_postgresql`) and is the text of the Phase 4A run.

The listing was checked against the stored requests. With the 30 data lines replaced by the placeholder, it equals the user message of 34,244 of 34,244 requests of the obfuscated gpt-6-luna input files (the dated-label arms differ only in the day labels). Each stored answer also carries the sha256 of the prompt that produced it in `detection_metadata.prompt_sha256`.

```
You are a market structure analyst specializing in dealer gamma positioning regimes.

TASK: Analyze this 30-day period and determine if it represents a PERSISTENT regime where dealer
↪ constraints create forced, directional flows.

## 30-DAY GEX DATA

{gex_data_table}

## REGIME CLASSIFICATION FRAMEWORK

### PERSISTENT REGIMES (Detect These)

**1. PERSISTENT POSITIVE REGIME**
- Definition: Dealers are LONG gamma, forced to sell into strength
- Criteria:
  * >70% of days (21+/30) have positive net GEX
  * Average magnitude >$5B
  * ≤5 sign flips across 30 days
  * Stable directional constraint

**Mechanism**: When dealers hold long gamma:
- Price rises → Dealers MUST sell shares (rebalance)
- Price falls → Dealers MUST buy shares (rebalance)
- Creates dampening, mean-reverting flows
- Constraint is STRUCTURAL (dealers cannot avoid)

**2. PERSISTENT NEGATIVE REGIME**
- Definition: Dealers are SHORT gamma, forced to buy into strength
- Criteria:
  * >70% of days (21+/30) have negative net GEX
  * Average magnitude >$5B
  * ≤5 sign flips across 30 days
  * Stable directional constraint

**Mechanism**: When dealers hold short gamma:
- Price rises → Dealers MUST buy shares (chase)
- Price falls → Dealers MUST sell shares (chase)
- Creates amplifying, momentum flows
- Constraint is STRUCTURAL (dealers cannot avoid)

---
```

```
781
782 ### NON-REGIMES (Reject These)
783
784 **3. TRANSITIONAL (Reject)**
785 - Frequent sign flips between positive/negative GEX
786 - No dominant regime direction (less than 70% same sign)
787 - Market in regime change period
788 - Example: 15 positive days, 15 negative days (50/50 split)
789
790 **Why Reject**: No persistent constraint. Dealers face mixed conditions daily. Not a structural
791 ↪ regime.
792
793 **4. LOW CONVICTION (Reject)**
794 - Consistent sign BUT weak magnitude (<$5B average)
795 - Example: 25 days positive, avg $2B GEX
796 - Insufficient constraint to create persistent forced flows
797
798 **Why Reject**: Even if sign is consistent, magnitude too weak to force dealers into meaningful
799 ↪ positions. Not a structural constraint.
800
801 ---
802
803 ## ANALYSIS QUESTIONS
804
805 Systematically evaluate the 30-day window:
806
807 **Step 1: Sign Persistence**
808 1. Count days with positive net GEX
809 2. Count days with negative net GEX
810 3. Calculate persistence percentage: max(positive_days, negative_days) / 30 * 100
811 4. Does it meet 70% threshold (21+ days)?
812
813 **Step 2: Magnitude Assessment**
814 1. Calculate average GEX magnitude (absolute value): sum(|net_gex|) / 30
815 2. Is average magnitude ≥$5B?
816 3. Check for extreme outliers that might distort average
817
818 **Step 3: Stability Check**
819 1. Count sign flips: How many times does GEX switch from pos→neg or neg→pos?
820 2. Are there ≤5 sign flips across 30 days?
821 3. Stable regime should have low flip count
822
823 **Step 4: Regime Classification**
824 - If Steps 1, 2, 3 all pass AND positive dominates → PERSISTENT POSITIVE
825 - If Steps 1, 2, 3 all pass AND negative dominates → PERSISTENT NEGATIVE
826 - If Step 1 passes but Step 2 fails → LOW CONVICTION (reject)
827 - If Step 1 fails → TRANSITIONAL (reject)
828
829 ---
830
831 ## CONFIDENCE CALIBRATION (Mechanical Guidance)
832
833 Use these concrete anchors to calibrate confidence:
834
835 **90-100 (Very High Confidence)**
836 - 25-30 days same sign (83-100% persistence)
837 - Average magnitude >$10B
838 - 0-2 sign flips (highly stable)
839 - Example: "29 negative days, avg $15B, 1 flip"
840
841 **70-89 (High Confidence)**
842 - 21-24 days same sign (70-80% persistence)
843 - Average magnitude $5-10B
844 - 2-4 sign flips (moderately stable)
845 - Example: "23 negative days, avg $7B, 3 flips"
846
847 **50-69 (Borderline - Use with Caution)**
848 - 18-20 days same sign (60-67% persistence)
849 - Average magnitude $3-5B
850 - 5-7 sign flips
851 - Example: "20 negative days, avg $4B, 6 flips"
852 - **Note**: Borderline cases should generally be REJECTED unless other factors strengthen
853 ↪ confidence
854
855 **0-49 (Reject - Not Persistent)**
```

```
- <18 days same sign (<60% persistence)
- OR average magnitude <$3B
- OR >7 sign flips
- These are NOT persistent regimes

**Important**: Confidence is a FILTER, not a probability. Use it to distinguish clear regimes (70+) from borderline (50-69) from noise (<50).

---

## OUTPUT FORMAT (JSON)

Provide your analysis in this exact JSON structure:

```json
{
    "regime_detected": true/false,
    "regime_type": "persistent_positive|persistent_negative|transitional|low_conviction",
    "positive_days": <count as integer>,
    "negative_days": <count as integer>,
    "avg_magnitude_billions": <value as number>,
    "sign_flips": <count as integer>,
    "persistence_pct": <percentage as number>,
    "confidence": <integer 0-100>,
    "reasoning": "Explain step-by-step why this is/isn't a persistent regime. Reference specific metrics (persistence %, avg magnitude, sign flips). If rejecting, state which criterion failed."
}
```

IMPORTANT: All numeric fields (confidence, positive_days, negative_days, sign_flips, avg_magnitude_billions, persistence_pct) MUST be numbers (integers or decimals), NOT words like "thirty-five" or "fifty".

regime_detected Rules:
- `true` ONLY if regime_type is "persistent_positive" or "persistent_negative"
- `false` if regime_type is "transitional" or "low_conviction"

KEY PRINCIPLES

1. **Selectivity is Expected**: Most windows will NOT be persistent regimes (expect 30-50% detection rate)

2. **ALL Criteria Must Pass**: Persistence + Magnitude + Stability required for detection

3. **Rejection is Valid**: Saying "no persistent regime" is a correct answer for transitional/weak periods

4. **Mechanical Over Qualitative**: Use concrete thresholds (70%, $5B, 5 flips) rather than subjective judgment

5. **Structural Focus**: Only detect when dealers are FORCED into directional positions by constraints

Analyze the 30-day GEX data above and provide your regime classification in JSON format.
```

Appendix A.3. Output Schema

The answer is constrained by the following strict JSON schema, passed as `response_format`. With `strict` set, the model can neither omit nor add a field. The boolean `regime_detected` is the primary label of the study and `regime_type` the secondary one; the counts, the mean magnitude and the persistence are the model's own report and are compared with the values of the deterministic classifier (Section 4.10).

```
{
  "type": "json_schema",
  "json_schema": {
    "name": "regime_classification",
    "strict": true,
    "schema": {
```

```
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "regime_detected": {
          "type": "boolean"
        },
        "regime_type": {
          "type": "string",
          "enum": [
            "persistent_positive",
            "persistent_negative",
            "transitional",
            "low_conviction"
          ]
        },
        "positive_days": {
          "type": "integer"
        },
        "negative_days": {
          "type": "integer"
        },
        "avg_magnitude_billions": {
          "type": "number"
        },
        "sign_flips": {
          "type": "integer"
        },
        "persistence_pct": {
          "type": "number"
        },
        "confidence": {
          "type": "integer"
        },
        "reasoning": {
          "type": "string"
        }
      },
      "required": [
        "regime_detected",
        "regime_type",
        "positive_days",
        "negative_days",
        "avg_magnitude_billions",
        "sign_flips",
        "persistence_pct",
        "confidence",
        "reasoning"
      ]
    }
  }
}
```

Appendix A.4. Request Body

Each line of a batch input file is one request of the following form.

```
{
  "custom_id": "<arm>|<symbol>|<window_end>",
  "method": "POST",
  "url": "/v1/chat/completions",
  "body": {
    "model": "gpt-6-luna",
    "messages": [
      {
        "role": "user",
        "content": "<prompt text of Appendix A.2>"
      }
    ],
    "max_completion_tokens": 4000,
    "reasoning_effort": "medium",
    "response_format": {
      "type": "json_schema",
      "json_schema": "<schema of Appendix A.3>"
    }
```

```
  }
}
```

Appendix B. Series Definitions in SQL

The daily series of Table 1 are defined by the four read-only statements of Appendices B.1 to B.4, which are reproduced verbatim from the data audit of 2026-10-28 on the project database; Appendix B.5 defines the derived and recovered series. The comments and the file names in the usage lines are those of the audit files. The audit calls the archive table "frozen" and its convention "old"; Section 3.3 shows that this table is put-less from October 2023, so it is not a clean old convention (series key `old_bs`). In the statements `:sym` is the symbol (SPY, QQQ or IWM) and `:y0`, `:y1` bound one calendar year, because the raw table is read one symbol-year per statement. Every statement stops at 2026-10-09, the last day before the vendor switch (Section 3.3). The first two statements return the 30 daily values of every window. The pipeline stores the daily rows and cuts the windows in a second step (Section 3.4); unlike that step, the first statement drops a window when one of its 30 days of the `gex.options_daily_summary` calendar is missing from the archive table.

Appendix B.1. Pre-Rebuild Archive Table (`old_bs`)

```
-- SERIES (a) FROZEN old convention (read-only archive). usage: psql -v sym=SPY -f
↪ sql_series_a_frozen.sql
-- Calendar = NEW table calendar for the symbol; a 30-day window is valid only if all 30
↪ consecutive NEW-calendar days exist in the frozen table.
-- (frozen max trading_date = 2026-10-02, so the last window ends 2026-10-02; IWM/QQQ have no rows
↪  2020-12-03..2025-12-30)
WITH cal AS (SELECT trading_date, row_number() OVER (ORDER BY trading_date) AS rn
             FROM gex.options_daily_summary WHERE symbol = :'sym' AND trading_date <= DATE
↪ '2026-10-09'),
     f AS (SELECT c.rn, c.trading_date, z.total_gex, z.net_call_gex, z.net_put_gex, z.
↪ underlying_price
           FROM cal c JOIN backups.options_daily_summary_pre_rebuild_20261004 z
             ON z.symbol = :'sym' AND z.trading_date = c.trading_date)
SELECT e.trading_date AS window_end,
       array_agg(w.total_gex     ORDER BY w.rn) AS total_gex_30,
       array_agg(w.net_call_gex  ORDER BY w.rn) AS call_gex_30,
       array_agg(w.net_put_gex   ORDER BY w.rn) AS put_gex_30
FROM f e JOIN f w ON w.rn BETWEEN e.rn - 29 AND e.rn
GROUP BY e.trading_date, e.rn
HAVING count(*) = 30          -- hole-free window
ORDER BY e.rn;
```

Appendix B.2. Rebuilt Daily Table (`new_full`)

```
-- SERIES (b) NEW 0DTE-inclusive stored-gamma convention (gex.options_daily_summary, rebuilt
↪ 2026-10-08). usage: psql -v sym=SPY -f sql_series_b_new.sql
WITH s AS (SELECT trading_date, total_gex, net_call_gex, net_put_gex, underlying_price,
                  row_number() OVER (ORDER BY trading_date) AS rn
           FROM gex.options_daily_summary
           WHERE symbol = :'sym' AND trading_date <= DATE '2026-10-09')      -- alpha_vantage ends
↪  2026-10-09; cboe_delayed starts 2026-10-12
SELECT e.trading_date AS window_end,
       array_agg(w.total_gex    ORDER BY w.rn) AS total_gex_30,
       array_agg(w.net_call_gex ORDER BY w.rn) AS call_gex_30,
       array_agg(w.net_put_gex  ORDER BY w.rn) AS put_gex_30
FROM s e JOIN s w ON w.rn BETWEEN e.rn - 29 AND e.rn
WHERE e.rn >= 30
GROUP BY e.trading_date, e.rn
HAVING count(*) = 30
ORDER BY e.rn;
```

Appendix B.3. Raw Chains Without Same-Day Expiries (`new_ex0dte`)

```
-- SERIES (c) NEW ex-0DTE from raw chains. ONE symbol-year per call (never wider). usage:
```

```
--   psql -v sym=SPY -v y0=2024-01-01 -v y1=2025-01-01 -f sql_series_c_ex0dte.sql      (loop y0/y1 over years; SPY/IWM 2008..2026, QQQ 2011..2026)
-- enable_seqscan=off is a SESSION planner setting (no writes): the default plan seq-scans the whole yearly partition (~4.5 s warm / 8.9 s cold per symbol-year);
-- the (symbol, trading_date) index/bitmap plan takes 0.5-1.8 s and returns identical numbers (max rel diff 5e-13).
SET enable_seqscan = off;
SELECT trading_date,
       count(*)                                                    AS n_contracts,
       count(*) FILTER (WHERE expiration = trading_date)           AS n_0dte,
       sum(CASE WHEN option_type = 'call' THEN g END)  FILTER (WHERE expiration > trading_date) AS call_gex_ex0dte,
      -sum(CASE WHEN option_type = 'put'  THEN g END)  FILTER (WHERE expiration > trading_date) AS put_gex_ex0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END) FILTER (WHERE expiration > trading_date) AS total_gex_ex0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END) FILTER (WHERE expiration = trading_date) AS total_gex_0dte,
       sum(CASE WHEN option_type = 'call' THEN g ELSE -g END)
        AS total_gex_all  -- ties to gex.options_daily_summary.total_gex (<=2e-7 rel)
FROM (SELECT trading_date, option_type, expiration,
             gamma::float8 * open_interest::float8 * 100.0 * underlying_price::float8 * underlying_price::float8 * 0.01 AS g   -- float8: real*int would stay float4
      FROM gex.options_chains_partitioned
      WHERE symbol = :'sym' AND trading_date >= :'y0' AND trading_date < :'y1'
        AND trading_date <= DATE '2026-10-09'                     -- vendor cap: alpha_vantage ends 2026-10-09, cboe_delayed starts 2026-10-12
        AND gamma IS NOT NULL AND open_interest > 0 AND underlying_price > 0) t
GROUP BY trading_date
ORDER BY trading_date;
```

Appendix B.4. Late-Session Intraday Snapshot (`intraday_late`)

The statement is the one tested in the audit and returns Variant 1 of its comments, the `market_close` slot, which equals the end-of-day row. The series of the study is Variant 2: the filter on the clock time is replaced by a filter on the snapshot type (`snapshot_type` equal to `final_rush`, 19:55:59 UTC), so that the snapshot is selected by type and not by clock time, and the three irregular days 2026-07-29, 2026-08-03 and 2026-09-14 are dropped when the series is built.

```
-- SERIES (d) INTRADAY late-session snapshot, SPY. Valid for 2026-07-27..2026-10-09 (EDT: 20:00 UTC = 16:00 ET; re-derive the cut after 2026-11-01 DST end).
-- Variant 1 'late' = last snapshot with timestamp < 20:01:00 UTC  -> the market_close slot (20:00:59), which EQUALS the daily summary row (not independent).
-- Variant 2 (independent of the EOD row): add  AND snapshot_type = 'final_rush'  (19:55:59 UTC).
SELECT DISTINCT ON (d)
       d AS trading_date, snapshot_timestamp, snapshot_type, total_gex, net_call_gex, net_put_gex, underlying_price, total_contracts
FROM (SELECT *, (snapshot_timestamp AT TIME ZONE 'America/New_York')::date AS d
      FROM gex.intraday_gex_summary
      WHERE symbol = 'SPY'
        AND snapshot_timestamp >= TIMESTAMPTZ '2026-07-27 00:00+00' AND snapshot_timestamp < TIMESTAMPTZ '2026-10-10 00:00+00'
        AND snapshot_type <> 'test'
        AND (snapshot_timestamp AT TIME ZONE 'UTC')::time < TIME '20:01:00') x
ORDER BY d, snapshot_timestamp DESC;
```

Appendix B.5. Derived and Recovered Series

Call leg only (`call_only`). The sum of the call terms g of the third statement over all expiries, same-day expiries included, with the put terms dropped. In the rebuilt summary table this is `net_call_gex` (up to the float4 rounding of the stored column), which the second statement returns as `call_gex_30`.

Scaled series (`new_full_scaled`). The `new_full` values of QQQ and IWM multiplied by one fixed factor per symbol, which the following statement computes (tested read-only on 2026-10-29). The factor and the number of days are stored with the series.

```
-- new_full_scaled: factor of one symbol = median over 2020-01-01..2026-10-09 of the ratio of SPY's open-interest notional to the symbol's.
```

```
-- usage: psql -v sym=QQQ     (QQQ or IWM; the scaled values are the new_full values of the symbol
↪  times this factor)
WITH n AS (SELECT symbol, trading_date, total_weighted_oi::float8 * 100.0 * underlying_price::
↪ float8 AS notional
           FROM gex.options_daily_summary
           WHERE symbol IN ('SPY', :'sym') AND trading_date BETWEEN DATE '2020-01-01' AND DATE
↪ '2026-10-09')
SELECT percentile_cont(0.5) WITHIN GROUP (ORDER BY s.notional / q.notional) AS factor,
       count(*) AS n_days
FROM n s JOIN n q ON q.trading_date = s.trading_date AND s.symbol = 'SPY' AND q.symbol = :'sym'
WHERE q.notional > 0;
```

Recovered input series. The series `jan2026_phase4a`, `p3_2024` and `p4_2020` (and the dated twins of the last two) are not defined by SQL. They are the daily values parsed from the data lines of the surviving batch input files `phase4a_postgresql_20260104_184832.jsonl`, `phase3_full_2024_baseline_v2.jsonl` and `batch_regime_20251120_104630.jsonl` (folder `reports/validation/paper2_regime_windows/batch_jobs/`); overlapping windows are checked for consistent values (Section 4.1).

Appendix C. Per-Year Tables

Tables A1–A4 give the numbers behind the figures of Section 4 for every year, taken from the files that `scripts/validation/electronics/05_report.py` writes to `docs/papers/electronics/results/`. The years are the calendar years of the window end, and rates are percentages of parsed windows. Wilson and quarter-block bootstrap intervals accompany every rate in `detection_by_year.csv` and `stats.yaml`; the Wilson intervals treat overlapping windows as independent and are optimistic.

Table A1. Mean absolute daily GEX (USD billion) by series and year, SPY. Each mean is taken over the days that the series has in common with `new_full`; the `new_full` column and the day count use the days shared with `new_ex0dte`. A dash marks a year outside the series.

Year	Days	jan2026_phase4a	call_only	old_bs	new_ex0dte	new_full
2008	253	–	–	1.39	1.59	1.60
2009	250	–	–	0.98	0.88	0.88
2010	251	–	–	1.01	1.14	1.16
2011	250	–	–	1.79	1.99	2.02
2012	250	–	–	1.54	1.71	1.78
2013	252	–	–	2.34	2.29	2.51
2014	252	–	–	3.03	3.07	3.36
2015	252	–	–	2.69	2.91	3.37
2016	250	–	–	3.99	3.76	4.23
2017	240	–	–	6.43	5.82	6.53
2018	249	–	–	4.79	4.63	5.32
2019	251	–	–	3.62	3.39	4.11
2020	252	3.24	10.51	2.76	2.57	3.24
2021	252	4.97	17.13	3.98	3.50	4.99
2022	251	5.48	9.27	4.42	4.62	5.46
2023	250	10.62	17.04	8.39	4.59	6.10
2024	252	20.19	20.19	16.19	4.13	6.46
2025	250	19.21	19.21	14.71	4.55	7.13
2026	193	–	–	6.00	5.76	8.37

Table A2. Detection rate of the deterministic classifier by series and window-end year, SPY, in percent of windows. Windows are the windows of `new_full`. A dash marks a year without windows in the series.

Year	Windows	`jan2026_phase4a`	`call_only`	`old_bs`	`new_ex0dte`	`new_full`
2008	224	–	–	0.0	0.0	0.0
2009	250	–	–	0.0	0.0	0.0
2010	251	–	–	0.0	0.0	0.0
2011	250	–	–	0.0	0.0	0.0
2012	250	–	–	0.0	0.0	0.0
2013	252	–	–	0.0	0.0	0.0
2014	252	–	–	13.5	8.3	9.1
2015	252	–	–	0.4	1.6	7.9
2016	250	–	–	26.4	22.0	28.0
2017	240	–	–	38.8	19.6	32.1
2018	249	–	–	32.1	23.7	34.5
2019	251	–	–	7.2	3.2	10.0
2020	252	12.1	90.6	16.7	10.7	22.2
2021	252	4.0	100.0	3.6	0.0	3.6
2022	251	32.3	100.0	26.3	36.7	40.6
2023	250	19.3	100.0	22.5	25.2	22.4
2024	252	100.0	100.0	100.0	13.1	10.3
2025	250	100.0	100.0	100.0	16.4	13.6
2026	193	–	100.0	23.5	21.2	11.4

Table A3. Detection rate of gpt-6-luna by arm and window-end year, in percent of parsed windows (a dash marks a year without windows in the arm). The columns are the arms `full_spy`, `ex0dte_spy`, `full_qqq`, `full_qqq_scaled`, `ex0dte_qqq`, `full_iwm`, `full_iwm_scaled` and `ex0dte_iwm` of Table 3 (full = `new_full`, scaled = `new_full_scaled`, ex-0DTE = `new_ex0dte`); the bridge, intraday and Phase 3/4 arms are reported in Sections 4.3, 4.9 and 4.11.

	SPY		QQQ			IWM		
Year	full	ex-0DTE	full	scaled	ex-0DTE	full	scaled	ex-0DTE
2008	0.0	0.0	–	–	–	0.0	–	–
2009	0.0	0.0	–	–	–	0.0	–	–
2010	0.0	0.0	–	–	–	0.0	–	–
2011	0.0	0.0	0.0	–	–	0.0	–	–
2012	0.0	0.0	0.0	–	–	0.0	–	–
2013	0.0	0.0	0.0	–	–	0.0	–	–
2014	9.1	8.3	0.0	–	–	0.0	–	–
2015	7.9	1.6	0.0	–	–	0.0	–	–
2016	27.6	22.0	0.0	–	–	0.0	–	–
2017	31.7	19.2	0.0	–	–	0.0	–	–
2018	34.9	22.9	0.0	–	–	0.0	–	–
2019	10.0	3.2	0.0	–	–	0.0	–	–
2020	22.2	10.7	0.0	0.0	0.0	0.0	9.9	0.0
2021	3.6	0.0	0.0	23.8	0.0	0.0	65.9	0.0
2022	41.8	37.1	0.0	61.0	0.0	0.0	100.0	0.0
2023	23.2	26.0	0.0	17.6	0.0	0.0	79.2	0.0
2024	11.1	14.3	0.0	8.4	0.0	0.0	20.6	0.0
2025	14.0	16.4	0.0	20.4	0.0	0.0	51.2	0.0
2026	11.4	21.2	0.0	16.1	0.0	0.0	45.1	0.0

Table A4. Run ledger: one row per Batch job, with the token usage and the cost priced at Batch rates. The identifier has the form `<arm>_<model>_<mode>_<UTC timestamp>_c<chunk>`. Only runs paid in this study enter the total; the synchronous pilot (`sync`) is included in it. The three backfilled o4-mini runs are listed below it, as stored answers of earlier runs that are not spend of this study.

Run identifier	Requests	Prompt tokens	Completion tokens	Cost (USD)	Status
`bridge_jan_gpt-6-luna_batch_20261029T203742Z_c0`	1,476	2,823,588	1,086,712	0.41	ingested
`bridge_jan_gpt-6-luna_batch_20261029T203742Z_c1`	1,476	2,823,588	1,082,280	0.41	ingested
`full_spy_gpt-6-luna_batch_20261029T203745Z_c0`	1,227	2,347,251	871,277	0.34	ingested
`full_spy_gpt-6-luna_batch_20261029T203745Z_c1`	1,226	2,345,338	969,842	0.36	ingested
`full_spy_gpt-6-luna_batch_20261029T203745Z_c2`	1,226	2,345,338	1,004,850	0.37	ingested
`full_spy_gpt-6-luna_batch_20261029T203745Z_c3`	1,226	2,345,338	1,024,540	0.37	ingested
`ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c0`	1,227	2,347,251	839,753	0.33	ingested
`ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c1`	1,226	2,345,338	968,965	0.36	ingested
`ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c2`	1,226	2,345,338	951,505	0.36	ingested
`ex0dte_spy_gpt-6-luna_batch_20261029T203749Z_c3`	1,226	2,345,338	982,560	0.36	ingested
`p34_obf_2024_gpt-6-luna_batch_20261029T203752Z_c0`	234	447,642	156,636	0.06	ingested
`p34_dated_2024_gpt-6-luna_batch_20261029T203754Z_c0`	234	461,682	154,330	0.06	ingested
`p34_obf_2020_gpt-6-luna_batch_20261029T203757Z_c0`	234	447,642	132,619	0.06	ingested
`p34_dated_2020_gpt-6-luna_batch_20261029T203759Z_c0`	234	461,682	130,805	0.06	ingested
`full_qqq_gpt-6-luna_batch_20261029T204029Z_c0`	1,354	2,590,202	1,012,074	0.38	ingested
`full_qqq_gpt-6-luna_batch_20261029T204029Z_c1`	1,353	2,588,289	1,004,705	0.38	ingested
`full_qqq_gpt-6-luna_batch_20261029T204029Z_c2`	1,353	2,588,289	1,057,334	0.39	ingested
`full_iwm_gpt-6-luna_batch_20261029T204033Z_c0`	1,228	2,349,164	762,229	0.31	ingested

Table A5. Run ledger (continued): remaining Batch jobs, the total of this study and the backfilled runs.

Run identifier	Requests	Prompt tokens	Completion tokens	Cost (USD)	Status
full_iwm_gpt-6-luna_batch_20261029T204033Z_c1	1,228	2,349,164	719,759	0.30	ingested
full_iwm_gpt-6-luna_batch_20261029T204033Z_c2	1,228	2,349,164	826,909	0.32	ingested
full_iwm_gpt-6-luna_batch_20261029T204033Z_c3	1,228	2,349,164	855,981	0.33	ingested
full_qqq_scaled_gpt-6-luna_batch_20261029T204036Z_c1	877	1,677,701	750,362	0.27	ingested
full_qqq_scaled_gpt-6-luna_batch_20261029T204036Z_c0	877	1,677,701	694,967	0.26	ingested
full_iwm_scaled_gpt-6-luna_batch_20261029T204039Z_c0	878	1,679,614	614,557	0.24	ingested
full_iwm_scaled_gpt-6-luna_batch_20261029T204039Z_c1	877	1,677,701	695,367	0.26	ingested
ex0dte_qqq_gpt-6-luna_batch_20261030T010447Z_c1	892	1,706,396	680,360	0.26	ingested
ex0dte_qqq_gpt-6-luna_batch_20261030T010447Z_c0	892	1,706,396	622,593	0.24	ingested
ex0dte_iwm_gpt-6-luna_batch_20261030T010451Z_c0	893	1,708,309	560,829	0.23	ingested
ex0dte_iwm_gpt-6-luna_batch_20261030T010451Z_c1	892	1,706,396	616,040	0.24	ingested
intraday_spy_gpt-6-luna_batch_20261029T204042Z_c0	54	103,302	45,525	0.02	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c0	1,218	2,330,034	813,090	0.32	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c1	1,218	2,330,034	970,974	0.36	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c2	1,217	2,328,121	953,927	0.35	ingested
bridge_old_gpt-6-luna_batch_20261030T010455Z_c3	1,217	2,328,121	830,288	0.32	ingested
pilot_bridge_jan_gpt-6-luna_sync_20261029T203454Z_c0	40	76,520	28,015	0.02	ingested
Total (this study)	34,712	66,432,136	25,472,559	9.70	
Backfilled o4-mini runs (not spend of this study)					
r1_o4mini_phase4a_20260104	1,476	2,655,324	2,928,992	7.90	backfill
r1_p3_o4mini_20251120	223	414,557	405,127	1.12	backfill
r1_p4_o4mini_20251120	223	414,557	325,436	0.94	backfill

Appendix D. Data Availability Details

The repository holds the condensed material and the results. The condensed files are in `data/backups/gex_llm/condensed/`.

- `detections.csv.gz`: 36,634 rows, one per stored model answer, with the verdict, the reported metrics and the parse status but without the reasoning or the raw text. The key is `arm`, `series_key`, `symbol`, `window_end`, `model`, `prompt_version`.
- `windows.csv.gz`: 38,501 rows, one per 30-day window, with the labels, the sha256 of the input as sent and the verdict of the deterministic rule but without the sequences.
- `series_index.csv`: 16 rows, one per series and symbol, with the row count, the first and last date, the source table and the hash of the builder.

Every number of the paper can be reproduced from the first two files (join on `series_key`, `symbol` and `window_end`) and the files of `docs/papers/electronics/results/`. The provenance of every Batch job (batch and file identifiers, sha256 of the input file, token counts and cost) is in `data/backups/gex_`

`llm/tables/runs/` and in the `manifest.json` and `batch_<k>.json` files of each run folder. The raw request and answer files (`input_<k>.jsonl`, `output_<k>.jsonl`) and the full tables are kept on the authors' machine and in the database; they are not part of the repository.

The full tables are loaded into four tables and one view of the schema `backups` of the research database: `gex_llm_regime_series` (the number shown to the model, one row per series, symbol and trading date), `gex_llm_regime_windows` (the 30-day input as sent, its sha256 and the verdict of the rule), `gex_llm_regime_runs` (one row per Batch job, pilot or backfilled run) and `gex_llm_regime_detections` (one row per model answer), with the view `gex_llm_regime_rates_summary` (yearly rate of the model next to the rate of the rule, by arm). The parquet copies of these four tables are the release format named in the Data Availability Statement. The options data are subject to the vendor's licence and are not redistributed.

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Abbreviations

The following abbreviations are used in this manuscript:

0DTE	Zero Days to Expiration
EOD	End of Day
GEX	Gamma Exposure
LLM	Large Language Model
OI	Open Interest

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